

Basic Concepts of Math

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Note to the Reader

Proofs, not computations, form the core of pure mathematics. Instead of focusing on techniques for solving problems, mathematicians focus on understanding abstract concepts well enough to prove theorems about them. With hard work, this text will help you become familiar with the rigorous language of mathematics and be able to write short proofs of your own. The text's main focus is the detailed examples, notes, and activities for learning proof techniques. The text also covers famous theorems, such as irrationality of $\sqrt{2}$, countability of the rational numbers, uncountability of the real numbers, the fundamental theorem of arithmetic, Fermat's little theorem, and the limit laws. The problems, which are carefully crafted to be thought-provoking, highlight additional important mathematical ideas, including a definition of addition and multiplication of integers, a paradox in set theory, winning strategies in two-player games, construction of the rational numbers, a definition of cardinality, and construction of the real numbers.

This text is intended for the mid-level course Math 2111: Basic Concepts of Math, taught by the author at Temple University in Fall 2025. The text is for your individual use only; please do not distribute it without the author's consent. Math 2111 should prepare you for more advanced mathematics courses, such as real analysis, modern algebra, number theory, and theoretical linear algebra, where fluency in the language of proofs is largely assumed. Moreover, some of the topics we will cover—sets, functions, and relations—are fundamental to all areas of mathematics but not taught in depth in other math courses.

Chapter 1

Mathematics

1.1 Axioms

Mathematics begins with statements assumed to be true, called **axioms** or **postulates**. For example, the axioms of plane geometry, which you can look up online. Our starting point is two axioms defining the integers and real numbers.

1.1.1 Axiom. There is a set $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, called the **integers**, with the usual operations of addition (+) and multiplication ($\cdot$), satisfying the following properties for all integers a, b, c :

$$(a + b) + c = a + (b + c) \quad \text{associativity of addition} \quad (1.1.2)$$

$$a + b = b + a \quad \text{commutativity of addition} \quad (1.1.3)$$

$$a + (-a) = 0 \quad \text{additive inverses} \quad (1.1.4)$$

$$a + 0 = a \quad \text{additive identity} \quad (1.1.5)$$

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{associativity of multiplication} \quad (1.1.6)$$

$$a \cdot b = b \cdot a \quad \text{commutativity of multiplication} \quad (1.1.7)$$

$$a \cdot 1 = a \quad \text{multiplicative identity} \quad (1.1.8)$$

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{distributive property} \quad (1.1.9)$$

1.1.10 Activity. Use the properties in 1.1.1 to prove the following.

(a) Let a, b, c be integers. Then $(a + b) \cdot c = a \cdot c + b \cdot c$.

(b) Let a be an integer. Then $a \cdot 0 = 0$.

(Hint: use 1.1.4, 1.1.5, and 1.1.9.)

(c) Let a be an integer. Then $a \cdot (-1) = -a$.

(Hint: use part (b) to show that $a \cdot (-1) + a = 0$.)

1.1.11 Axiom. There is a set $\mathbb{R}$, called the **real numbers**, with the usual operations of addition (+) and multiplication ($\cdot$), satisfying the same properties as the integers plus

one additional property: for every nonzero real number a , there is a real number $\frac{1}{a}$ such that $a \cdot \frac{1}{a} = 1$. We call $\frac{1}{a}$ the “multiplicative inverse” of a .

1.1.12 Note.

- For both $\mathbb{Z}$ and $\mathbb{R}$, subtraction is just addition of a negative number.
- Division in $\mathbb{R}$ is just multiplication by a multiplicative inverse, but this definition of division does not work for $\mathbb{Z}$ because the only integers that have multiplicative inverses are 1 and -1 . See 4.3.1 for a different definition of division.

Technically, the axioms of $\mathbb{Z}$ and $\mathbb{R}$ provided in 1.1.1 and 1.1.11 are not well-defined as they depend on the notion of “set” and don’t specify how the operations of addition and multiplication are defined. A rigorous description of the integers and real numbers is complicated and would distract from our goal of exploring a range of basic concepts in mathematics at this early stage. See 3.1.1 for a definition of “set” and Problems 1 and 18 for explorations related to the definition of integers and real numbers.

1.2 Definitions

All we have so far are two sets of “numbers” with some operations and properties. We now use **definitions** to introduce new mathematical concepts. In order to be useful, definitions have to be unambiguous, so they have to be stated very precisely.

Let’s try to define what it means for an integer to be “even”. The way we normally state a definition is:

Definition. An integer n is **even** if ...

We could finish the sentence with “ n is in the set $\{\dots, -4, -2, 0, 2, 4, \dots\}$ ”, but that would not be very enlightening. A better definition would describe “evenness” in terms of a property, such as “ n is divisible by 2”. But we have to be careful: we don’t know what it means to be “divisible by 2”, since that is not in our axioms, so we would have to first define “divisible”. Although we could do that, we can give a definition purely in terms of multiplication:

1.2.1 Definition. An integer n is **even** if there exists an integer k such that $n = 2k$.

While we’re at it, we might as well define “odd” too. One option would be to say “an integer n is odd if it is not even”, but there is a more illuminating definition:

1.2.2 Definition. An integer n is **odd** if there exists an integer k such that $n = 2k + 1$.

These definitions of even and odd are useful because they tell us something about how the integer decomposes into a sum and product of other integers. But it is not clear from our definitions that an integer cannot be both even and odd, so that is a fact we will have to prove in the next chapter.

A familiar concept we have not yet defined is “rational number”. If we try to define “rational number” using just the integers—“a rational number is one integer divided by another integer”—we run into problems because division isn’t defined for integers. We would have to build the rational numbers as a new set of numbers, define operations, and show that the usual properties hold. But since we are already assuming the existence of the real numbers, we can define a rational number as a special kind of real number. Then we can use division since it is defined for nonzero real numbers; see 1.1.12.

1.2.3 Definition. A real number r is **rational** if there exist integers a and b with $b \neq 0$ such that $r = a/b$.

We’d also like to define “irrational”. Can you think of a useful property that characterizes when a real number is irrational? There does not seem to be such a property, so we have to settle for:

1.2.4 Definition. A real number that is not rational is called **irrational**.

We’ll introduce many new definitions in later chapters. But first we introduce some other key notions in mathematics.

1.3 Theorems

Theorems are collections of mathematical statements that have been proven to be true. For instance, here’s a famous theorem you have surely encountered:

Theorem (Pythagoras). *Let a and b denote the lengths of the legs of a right triangle, and let c be the length of the hypotenuse. Then*

$$a^2 + b^2 = c^2.$$

We’ll focus on proofs in the next chapter. For now, let’s try to understand how theorems are stated:

- The “let ...” sentence establishes the context and notation: we’re dealing with a right triangle with side lengths a , b , and c . Since the sentence does not specify a particular right triangle, the theorem applies to *every* right triangle. That’s why we should think of a theorem as a “collection” of mathematical statements.
- The “then ...” sentence makes a statement about an equation those lengths satisfy. It is remarkable that this equation holds for *every* right triangle. This is the part of the theorem that requires a proof.

The word “theorem” is usually reserved for the most important theorems. Less important theorems are called **propositions**. A **lemma** is a theorem whose main purpose is to be used in the proof of a more important theorem. A **corollary** is a theorem whose proof follows almost immediately from a previous theorem.

Mathematical statements that have not yet been proven or disproven are called **claims** or **conjectures**. We will call less important mathematical statements “claims”, even if we then go on to prove them, in which case we can refer to them as **true claims**. Let’s take a look at two simple claims.

Claim. *Let n be an integer. If n is odd, then n^2 is odd.*

Again, let’s figure out what the claim is saying:

- The “let ...” sentence establishes notation: we are using the variable n to represent an integer. Since the sentence does not put any additional conditions on n , the claim applies to all integers.
- The second sentence is in the form “if ... then ...”. Roughly speaking, the “if” part imposes an additional condition on n , namely that “ n is odd”. Given this additional condition, the claim says that n^2 must be odd.

Another common sentence type is “... if and only if ...”:

Claim. *Let n be an integer. Then n is odd if and only if n^2 is odd.*

We’ll analyze the meaning of sentences containing “if, then” or “if and only if” in the next section. To do so, we will define “statement”, which we have already used to define “theorem”.

1.4 Statements

A (mathematical) **statement** is a meaningful sentence about mathematics that is either true or false, not both. We often denote statements by lower-case letters like p and q . The **truth value** of a statement is either true (T) or false (F).

1.4.1 Example.

- “2 is even” is a true statement.
- “ $1 + 2 = -7$ ” is a false statement.
- “ $1 = + + 3 -$ ” is not a statement because it is gibberish.

When we use p to denote a statement, the statement “ p ” has the same meaning as the statement “ p is true”, so we usually suppress the “is true”. For instance, “2 is even” has the same meaning as “2 is even is true”.

1.4.2 Activity.

- (a) Come up with your own examples of a true statement, a false statement, and a sentence that is not a statement.

- (b) Determine the truth value of the sentence “This sentence is false.” Is the sentence a statement?

Given two statements p and q , there are various ways to combine them to get a new statement.

- **“Not”**: The statement “it is not true that p ”, which has the same meaning as “ p is false”, is true when p is false and false when p is true. This statement is often shortened to “not p ” and can also be called the “negation of p ”.
- **“And”**: The statement “ p and q ” is true when both p and q are true and false otherwise.
- **“Or”**: The statement “ p or q ” is true when p or q (possibly both) are true and false when both p and q are false.
- **“If...then”**: The statement “if p , then q ” or “ p implies q ” is often denoted $p \implies q$. It is trivially true when p is false, since then the “if p ” condition is not met, and true when both p and q are true. It is false only when p is true and q is false.
- **“If and only if”**: The statement “ p if and only if q ”, which means “(if p , then q) and (if q , then p)”, is often denoted $p \iff q$. It is true when p and q have the same truth value and false otherwise.

1.4.3 Example.

- “It is not true that 2 is even”, which can be shortened to “2 is not even”, is a false statement. “It is not true that $1 + 2 = -7$ ”, which can be shortened to “ $1 + 2 \neq -7$ ”, is a true statement.
- “2 is even and $1 + 2 = -7$ ” is a false statement because “ $1 + 2 = -7$ ” is false.
- “2 is even or $1 + 2 = -7$ ” is a true statement because “2 is even” is true. “2 is even or $1 + 2 \neq -7$ ” is also a true statement.
- “If 2 is even, then $1 + 2 = -7$ ” is false, because “2 is even” is true and $1 + 2 = -7$ is false.
- “If 2 is even, then $1 + 2 \neq -7$ ” is true.
- “If 2 is not even, then $1 + 2 = 7$ ” is (trivially) true because “2 is not even” is false.
- “2 is even if and only if $1 + 2 = -7$ ” is a false statement.

1.4.4 Note. Here are some comments about how our notions of “or”, “if...then”, and “if and only if” relate to everyday usage.

- In everyday language, “or” can be ambiguous. I might say “I will buy peaches or I will buy bananas”, with the implication that I will buy one or the other but not both. This is *not* how “or” works in mathematics. The statement “ p or q ” is considered to be *true* when both p and q are true.
- A parent might say to their child: “If you get an ‘A’ in astrophysics, then I will buy you a bicycle”. When has the parent upheld their word? If the child gets an ‘A’ and the parent buys the bike, then all is well. If the child fails to get an ‘A’, then the parent need not buy a bike, though they might get it anyway as consolation. But if the child gets the ‘A’ and the parent does not buy the bicycle, then the child has a reason to be upset, and we would say the parent has been dishonest.
- Suppose I tell you “I will move to South Korea if and only if you will move to South Korea”. I am claiming that we will either move together or not at all, and I will be wrong if one of us goes while the other stays.

We can use a **truth table** to summarize the truth values of more complicated statements in terms of the truth values of their building blocks.

p	q	not p	p and q	p or q	if p , then q	p if and only if q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Here p and q denote statements, each of which is true or false. The four rows correspond to the four possible combinations of truth values for p and q , which appear in the first two columns. To determine the truth values in the other five columns, use the truth values for p and q and the definition of “not”, “and”, “or”, and so on.

Let p and q be statements. The **converse** of the statement “if p , then q ” is the statement “if q , then p ”. The **inverse** of “if p , then q ” is the statement “if not p , then not q ”. The **contrapositive** of “if p , then q ” is “if not q , then not p ”.

1.4.5 Activity. Let p and q be statements. Use truth tables to justify each statement.

- “If p , then q ” may not have the same truth value as its converse.
- “If p , then q ” may not have the same truth value as its inverse.
- “If p , then q ” always has the same truth value as its contrapositive.
- “Not (p and q)” always has the same truth value as “not p or not q ”. (The parentheses indicate that the “not” applies to the whole statement “ p and q ”.)
- “Not (p or q)” always has the same truth value as “not p and not q ”.

Parts (d) and (e) are called De Morgan's laws. They say that you can distribute “not” over an “and” or an “or”, but you have to flip “and” to “or” and vice versa.

Because of (c), we say that an “if...then” statement and its contrapositive are **logically equivalent**. Logical equivalence is the same notion as “if and only if”. Another way of saying “ p if and only if q ” is saying “ p is equivalent to q ”.

Now we are ready to analyze some claims.

- **Claim.** *Let n be an integer. If n is odd, then n^2 is odd.*

The “let ...” sentence establishes that n is an integer, so as we saw in 1.3, the claim really applies to all integers. When n is not odd, the “if...then” sentence is trivially true since it does not promise anything. The only way the “if...then” sentence can be false is if there is an integer n such that n is odd and n^2 is not odd.

- **Claim.** *Let n be an integer. Then n is odd if and only if n^2 is odd.*

By the definition of “if and only if”, the claim says that when n is odd, so is n^2 , and when n^2 is odd, so is n . Since the statement “if p , then q ” is equivalent to its contrapositive, “if n^2 is odd, then n is odd” is equivalent to “if n is not odd, then n^2 is not odd”.

- **Claim.** *Let m and n be integers. If mn is even, then m is even or n is even.*

The “let ...” sentence establishes that m and n are two integers, possibly the same integer, so this claim applies to all pairs of integers. By definition of “if...then”, the claim is true unless mn is even and “ m is even or n is even” is false, which happens when “ m is not even and n is not even” is true.

1.5 Problems

Problem 1. One way to define addition and multiplication for integers is as follows. Think of the integers as a list

$$\dots, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, \dots$$

that extends indefinitely to the left and to the right, allowing for unlimited counting in both directions. If a, b are integers, then $a + b$ is the integer obtained by starting at a and *counting on* by b (if b is positive, count on to the right; if b is negative, count on to the left). For example, $3 + (-5)$ means starting at 3 and moving 5 steps left, resulting in -2 , so $3 + (-5) = -2$.

With addition defined, multiplication can be defined as repeated addition: $a \cdot b$ means starting at 0 and counting on by a a total of b times (if b is negative, also change the sign of the result). For example, $(-2) \cdot 3$ means $0 + (-2) + (-2) + (-2)$, which can be computed by starting at 0 and moving two steps left three times, resulting in -6 . Thus $(-2) \cdot 3 = -6$.

When proving properties about operations defined in terms of counting, it is useful to keep in mind the *fundamental principle of counting*: when counting a collection of objects, the result is independent of the order in which the objects are counted.

- (a) Using the “counting on” definition of addition, explain why the first four properties in 1.1.1 hold.

(Hint: one or two sentences per property should suffice. Be sure to explain the general case rather than giving examples involving specific numbers. For 1.1.2 and 1.1.3, you can view the equation as counting a certain collection of objects in two different ways and apply the fundamental principle of counting.)

- (b) Using the definition of multiplication as repeated addition, explain why the remaining four properties in 1.1.1 hold.

(Hint: again, be sure to explain the general case. For 1.1.7 and 1.1.9, think of counting a collection of objects arranged in a rectangle in two different ways. For 1.1.6, think of a collection of objects arranged in a box. It is okay to focus on the case where a, b, c are positive.)

Chapter 2

Proofs

2.1 Claims, revisited

As we saw in Chapter 1, axioms and definitions establish terminology, so they cannot be true or false. A theorem or claim, on the other hand, has a list of assumptions, called **premises**, and then makes an assertion, often indicated by a “then” statement, which we call its **conclusion**. For instance, consider the claim

Claim. *Let n be an integer. Assume n is odd. Then n^2 is odd.*

The sentence “let n be an integer” establishes notation, so it is a premise. The sentence “assume n is odd” is also a premise. The conclusion is “then n^2 is odd”. To write this claim more compactly, combine the premises into one sentence:

Claim. *Let n be an odd integer. Then n^2 is odd.*

This same claim can also be written using an “if...then”:

Claim. *Let n be an integer. If n is odd, then n^2 is odd.*

We don’t usually think of “If n is odd, then n^2 is odd” as the conclusion. Instead, we treat the “if” part as an additional premise and the “then” part as the conclusion. This works because an “if...then” sentence is trivially true when the “if” part is false, so the only interesting case is when the “if” part is true. Thus we treat the “if” part as a premise.

Another way to write the claim is:

Claim. *If n is an odd integer, then n^2 is odd.*

Usually it is easy to break down the claim into premises and the conclusion: just take the “then” statement to be the conclusion and everything else to be the premises. I like to write claims using “then” statements, but be aware that out in the wide world of mathematics, you could come across more confusing ways of writing the same claim:

Claim. *The square of every odd integer is odd.*

2.2 True claims

A **case** of a claim is a choice of values for all the variables in the claim that makes all the premises true. The cases of the above claim

Claim. *Let n be an odd integer. Then n^2 is odd.*

are exactly the choices of odd integer values for n , namely when $n = 1$, $n = -1$, $n = 3$, $n = -3$, and so on.

As another example, consider

Claim. *Let m and n be integers. Then $m + n$ is an integer.*

The cases of this claim are any choice of integer values for m and n , for instance $m = 0$ and $n = 1$, or $m = -3$ and $n = -3$, $m = 700$ and $n = 209$, and so on.

We say a claim is **true** or **valid** if the conclusion is true for every case of the claim. In other words, a claim is true if whenever all the premises are true, the conclusion is also true.

Often, a more useful way of understanding truth of a claim is through the notion of counterexamples. A **counterexample** is a case of a claim for which the conclusion of the claim is false. In this language, we can say

A claim is true if and only if it has no counterexamples.

Keep this sentence in mind as we move on to considering proofs.

2.3 What is a proof?

A (mathematical) **proof** is a short essay used to show that a claim is true. Although there are various ways to set up a proof, there is always a common goal: show the claim has no counterexamples.

While doing research, a mathematician often inadvertently comes up with claims that are false. To **disprove** a claim, namely to show a claim is false, find **one counterexample**. This is usually much easier than proving a claim is true. For a claim to be true, it has to be true in every possible case. A single counterexample makes a claim false, even if the claim is true in infinitely many other cases!

2.3.1 Example.

- **Claim.** *Let r be a real number. Then $r^2 > 0$.*

The claim is false because it fails when $r = 0$. A mathematician often tries to modify false claims to make them true. One way of doing this is:

- **Claim.** *Let r be a nonzero real number. Then $r^2 > 0$.*

This claim is true because the only counterexample of the previous claim has been removed due to the assumption “nonzero”. As you can see, claims must be stated very precisely. Some claims have a lot of assumptions that are needed to rule out counterexamples. By the way, why is $\sqrt{-1}$ not a counterexample?

Before discussing various techniques for proving claims, here is a rough idea of how to approach a claim you've never seen before, whose truth is unknown.

2.3.2 Note (Outline for Proving or Disproving Claims).

- *Step 1:* Before you attempt a proof, convince yourself that the claim is correct. Checking easy cases is often a good idea. See if you can find a counterexample. If you can, just write “This claim is false!”, state your counterexample, briefly explain why it is a counterexample, and you're done. If the claim has very few cases (a rare occurrence), try to check that the conclusion is true in each case (exhaustive proof). Otherwise move to Step 2.
- *Step 2:* On scratch paper, write down the premises at the top of the page, and the conclusion (which you are trying to deduce) at the bottom of the page. Think about what it would take to show the conclusion is true. Make sure you know what you want to show before you try to show it! (See 2.1 for tips on how to break down a claim into premises and a conclusion.)
- *Step 3:* Try to fill in the (both physical and logical) space between the premises and the conclusion. Use definitions, axioms (mainly arithmetic in our case), and previously proved theorems to build a chain of implications from the premises to the conclusion. (This is an attempt at a direct proof: if you get stuck, try a proof by contraposition or contradiction instead.)
- *Step 4:* Once you have an outline of the proof on your scratch paper, convert it into precise, crisp English sentences, possibly with some mathematical symbols mixed in. Label it “proof”, draw your favorite symbol at the end, and you have yourself a proof!

2.4 Exhaustive Proof

Rarely, a claim has just a few cases, for which the truth of the conclusion can easily be checked. For instance:

2.4.1 Claim. *Let n be an integer such that $1 \leq n \leq 4$. Then $(n + 1)^3 \geq 3^n$.*

Proof.

$$\text{For } n = 1, (1 + 1)^3 = 8 \geq 3 = 3^1.$$

$$\text{For } n = 3, (3 + 1)^3 = 64 \geq 27 = 3^3.$$

$$\text{For } n = 2, (2 + 1)^3 = 27 \geq 9 = 3^2.$$

$$\text{For } n = 4, (4 + 1)^3 = 125 \geq 81 = 3^4.$$

□

For an exhaustive proof, we clearly state each case and show that the conclusion is true in that case. That's all we have to write. We don't have to explain how exhaustive

proof works because it is a common technique, but make sure you understand why our proof of this claim is completely convincing.

Claims that are susceptible to exhaustive proof are usually too trivial to be of much interest. Interesting claims tend to have an infinite number of cases.

2.5 Direct proof

The most common kind of proof is a direct proof. For a direct proof, we build a chain of implications from the premises to the conclusion. In other words, we assume the premises are true and make deductions until we have deduced that the conclusion must be true. Thus the premises imply the conclusion, so there cannot be a counterexample.

Recall our definition of even and odd from Chapter 1:

Definition. An integer n is **even** if there exists an integer k such that $n = 2k$. An integer n is **odd** if there exists an integer k such that $n = 2k + 1$.

Let's apply the outline in 2.3.2 to

2.5.1 Claim. *Let n be an odd integer. Then n^2 is odd.*

Step 1: Checking some easy cases of the claim, we see that

- when $n = -3$, $(-3)^2 = 9$ is odd
- when $n = 1$, $1^2 = 1$ is odd
- when $n = -1$, $(-1)^2 = 1$ is odd
- when $n = 3$, $3^2 = 9$ is odd

so the claim seems plausible. Although it is easy to check individual cases of the claim, we cannot use exhaustive proof because there are infinitely many cases.

Step 2: Our premises state that n is an odd integer, namely that there is an integer k such that $n = 2k + 1$. The conclusion, which we want to deduce, is that n^2 is odd. What would it take to show that n^2 is odd? We have to *find* an integer j such that $n^2 = 2j + 1$.

Step 3: How do we find this integer j ? We need some way to use our information about n to deduce something about n^2 . We have an equation for n , so square it. This gives us

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, setting $j = 2k^2 + 2k$ yields $n^2 = 2j + 1$, so n^2 is odd. We've finished our scratch work; here is the proof:

Proof. Since n is odd, there exists an integer k such that $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is odd. □

2.5.2 Note.

- Notice how short and precise the finished product is. Since the claim tells us that n is an odd integer, we don't have to repeat that in the proof. We do write "since n is odd" because that is the reason why k exists. Think of writing a proof in analogy to writing an essay about a novel: don't restate premises (don't include pure plot summary), but do cite premises to justify your deductions (do talk about details from the novel in order to make interesting conclusions).
- We incorporate the computation into the second sentence and center the computation to make it more readable. We don't talk about an integer j because there is no need: the expression $2k^2 + 2k$ clearly defines an integer since addition and multiplication are operations on $\mathbb{Z}$. In general, it is better to avoid introducing new variables unless there is a good reason for doing so.
- We used the definition of "odd" twice: once to extract useful information about n (there exists k such that $n = 2k + 1$), and again to tell us what we need to prove (that there exists j such that $n^2 = 2j + 1$). If you don't know the definition of odd, you have no chance of proving a claim about odd integers. You have to know the main definitions so that you can prove claims involving them.

2.5.3 Activity. Prove each claim.

- Let n be an even integer. Then n^2 is even.
- Let n be an odd integer. Then n^3 is odd.

2.5.4 Note (Even and odd). Let n be an integer.

- If you know n is even, you can deduce that there is an integer k such that $n = 2k$.
- To *show* n is even you must *find* an integer j such that $n = 2j$.
- If you know n is odd, you can deduce that there is an integer k such that $n = 2k + 1$.
- To *show* n is odd you must *find* an integer j such that $n = 2j + 1$.
- To show n is not even, show it is odd (and vice versa). This works because of 2.7.5.

2.6 Proof by contraposition

Sometimes direct proofs are difficult because the premises don't easily get you to the conclusion.

2.6.1 Claim. *Let n be an integer. If n^2 is odd, then n is odd.*

Let's try to work out a direct proof. Our premises tell us that n is an integer and n^2 is odd, so there exists k such that $n^2 = 2k + 1$. To prove the conclusion, that n is odd, we have to find an integer j such that $n = 2j + 1$. But how can we use information about

n^2 to deduce something about n ? We might try taking the square root of both sides of the equation $n^2 = 2k + 1$, which yields the equation $n = \pm\sqrt{2k + 1}$. Unfortunately, this equation does not seem to help in finding j . What next?

This claim is easy to prove if we use a **proof by contraposition**. Recall that a statement of the form “if p , then q ” is equivalent to its contrapositive “if not q , then not p ”. So we can prove the contrapositive instead!

In our case, the contrapositive is the claim “Let n be an integer. If n is not odd, then n^2 is not odd.” We’ll prove shortly that any integer that is not odd is even. Granting that fact, the claim we want to prove is “Let n be an integer. If n is even, then n^2 is even.” The proof is similar to our proof of 2.5.1 above.

Proof by contraposition. Assume n is even. Then there exists an integer k such that $n = 2k$. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. Since $2k^2$ is an integer, n^2 is even. $\square$

Note that we started the proof by pointing out that we are using proof by contraposition. Then we began proving the contrapositive of the claim. We wrote “Assume n is even” instead of “Since n is even” to remind the reader about the assumption in the contrapositive, but we didn’t need to write down the full contrapositive. Of course, you should write down the contrapositive on your scratch paper.

2.6.2 Activity. Let n be an integer. Prove each claim by contraposition.

- (a) If n^2 is even, then n is even.
- (b) If n^3 is odd, then n is odd.

2.7 Proof by contradiction

Proof by contraposition only makes sense when the claim has a sentence of the form “if...then”. As we saw in 2.1, you can force a claim to have an “if...then” structure by choosing any premise, writing it after the “if”, and writing the conclusion after the “then”. But what if there is no premise, as in 2.7.4, below? And if there are many premises, which do you choose?

There is a powerful proof technique called **proof by contradiction**, which generalizes proof by contraposition. The idea of proof by contradiction is to rule out the possibility of a counterexample by supposing there were a counterexample, namely a case of the claim that makes the conclusion false, and then deducing a statement that is clearly false, called a **contradiction**. If this is achieved, then the counterexample cannot exist. Here is a simple example illustrating the technique.

2.7.1 Claim. Let n be an integer. If n^2 is even, then n is even.

Proof. Assume for contradiction that n is odd. Then there is an integer k such that $n = 2k + 1$. Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k$ is an integer, n^2 is odd, contradicting the fact that n^2 is even. $\square$

As usual, the way we write the proof suppresses the details of how proof by contradiction works. We begin the proof by writing just “Proof”, but the sentence “Assume for contradiction” informs the reader that this is a proof by contradiction, not a direct proof. What does the sentence “Assume for contradiction that n is odd” really mean? It means

“Assume the claim has a counterexample. For this case of the claim, n^2 is even, and the conclusion “ n is even” is false, so n is odd. We will now try to show that it is impossible for both n^2 to be even and n to be odd.”

To get the contradiction, we use the fact that n is odd to deduce that n^2 is odd, contradicting the fact that n^2 is even.

In summary: when n^2 is even, n cannot be odd. Why not? Because if n were odd, then we can easily show that n^2 would have to be odd, which is impossible since we are assuming that n^2 is even. Since n being odd would lead us to conclude something impossible, n cannot be odd. Thus n is even, which is what we were trying to prove.

Here is another example where a direct proof falls short. Since the statement is not of the form “if...then”, proof by contradiction is more natural than proof by contraposition.

2.7.2 Claim. *Let r be irrational. Then $2r$ is irrational.*

Why is a direct proof of this claim so difficult? Because to show that $2r$ is irrational, you would have to show that $2r$ cannot be written as a/b , where a and b are integers. It is often hard to directly show that something is not possible. Also, the premise “Let r be irrational” does not give you a good starting point for a proof. Proof by contradiction lets you assume that $2r$ is rational, which is much easier to work with.

Proof. Assume for contradiction that $2r$ is rational. Then there are integers a, b with $b \neq 0$ such that $2r = a/b$. But then $r = a/(2b)$, so r is rational, contradicting the fact that r is irrational. Thus $2r$ must be irrational. $\square$

2.7.3 Note. Proof by contradiction generalizes both proof by contraposition and direct proof! Indeed, proof by contraposition is the special case of proof by contradiction in which you use the negation of the conclusion to deduce that one of the premises is false, which contradicts that premise. And direct proof is the special case of proof by contradiction in which you assume the conclusion is false and then use the premises to prove the conclusion is true, contradicting the fact that the conclusion is false. This shows you that proof by contradiction is *more powerful* than both direct proof and proof by contraposition. Still, a direct proof is usually logically clearer than a proof by contradiction, so you should use a direct proof when possible.

Next, we’ll see a famous proof by contradiction that establishes the irrationality of the real number $\sqrt{2}$. A direct proof would be difficult since there is no obvious starting point due to the absence of premises! We’ll need to use the familiar fact that rational numbers can be written in “lowest terms”, which means that the numerator and denominator have no common factor greater than 1. We’ll also need to use 2.7.1, which states that for all integers n , if n^2 is even, then n is even.

2.7.4 Theorem. $\sqrt{2}$ is irrational.

Proof. Assume for contradiction that $\sqrt{2}$ is rational. Then there are integers a and b with $b \neq 0$ such that $\sqrt{2} = a/b$. Moreover, we may choose a and b to have no common factors larger than 1. Then $2 = a^2/b^2$, so $a^2 = 2b^2$. Thus a^2 is even, which implies that a is even. So there is an integer k such that $a = 2k$. Then $2b^2 = a^2 = (2k)^2 = 4k^2$, so $b^2 = 2k^2$. Thus b^2 is even, which implies that b is even. But we have shown that both a and b are even, so a and b each have a factor of 2, which contradicts the fact that a and b have no common factor larger than 1. $\square$

As promised, we will now give the proof that every integer is even or odd, but not both. Proof by contradiction will once again be useful.

2.7.5 Proposition. Let n be an integer. Then n is even or odd, but not both.

This claim is complicated. The conclusion really means “(n is even or odd) and (n is not even or not odd)”. Since this is an “and” statement, we have to prove each part. We’ll use a direct proof to show “ n is even or odd” and a proof by contradiction to show “ n is not even or not odd”.

Proof. First we prove that n is even or odd. Since we can write the set of integers as

$$\{\dots, -2, -1, 0, 1, 2, \dots\} = \{\dots, 2 \cdot (-1), 2 \cdot (-1) + 1, 2 \cdot 0, 2 \cdot 0 + 1, 2 \cdot 1, \dots\},$$

the integers alternate between even and odd, so n must be even or odd.

Next we prove that n is not even or not odd. Assume for contradiction that n is even and odd. Since n is even, $n = 2k$ for some integer k . Since n is odd, $n = 2j + 1$ for some integer j . Then $2k = 2j + 1$, so $2(k - j) = 1$. There are three possibilities for $k - j$:

- $k - j \geq 1$: then $2(k - j) \geq 2$, contradicting $2(k - j) = 1$.
- $k - j = 0$: then $2(k - j) = 0 \neq 1$.
- $k - j \leq -1$: then $2(k - j) \leq -2$, contradicting $2(k - j) = 1$.

In each case, there is a contradiction. Thus our assumption must be false. So n cannot be both even and odd, hence n is not even or not odd. $\square$

Note that we indicated which part of the conclusion we were proving. The first part was a direct proof, which is always assumed unless otherwise specified. For the second part, we made it clear that we were using proof by contradiction.

This proof was more challenging. It was not enough to just apply definitions; we had to use some ingenuity too. Don’t worry: most of the proofs you will write in this chapter will look more like the proofs of 2.5.1 and 2.6.1.

2.7.6 Activity. Prove each claim by contradiction.

- (a) Let r be irrational. Then $r + \frac{1}{2}$ is irrational.

(b) Let r be irrational. Then $r/3$ is irrational.

(c) $\sqrt{3}$ is irrational.

(Hint: Recall the definitions of rational and irrational. For (c), modify the proof of 2.7.4.)

We could also write (a) and (b) in the preceding activity as “if...then” statements, for instance “Let r be a real number. If r is irrational, then $r + \frac{1}{2}$ is irrational.” Then to set up a proof by contradiction, you assume the “then” statement is false. This is very similar to proof by contraposition.

2.8 Proving logical equivalence

Here is a claim involving logical equivalence.

2.8.1 Claim. *Let n be an integer. Then n is odd if and only if n^2 is odd.*

The claim is saying that the statements “ n is odd” and “ n^2 is odd” are logically equivalent. As mentioned in 1.4, an “if and only if” statement is two “if...then” statements combined. Another way of stating the above claim is “Let n be an integer. If n is odd, then n^2 is odd, and if n^2 is odd, then n is odd.” We denote the “If n is odd, then n^2 is odd” part of the conclusion by “ $\implies$ ” and the “If n^2 is odd, then n is odd” part by “ $\impliedby$ ”. To prove the conclusion, we have to prove both parts of the “and” statement. We’ve already proved each part in this chapter, so all we have to do is combine the two proofs and clarify which part we are proving by writing “ $\implies$ ” and “ $\impliedby$ ”.

Proof. $\implies$: Since n is odd, there exists an integer k such that $n = 2k + 1$. Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k$ is an integer, n^2 is odd.

$\impliedby$: Proof by contraposition. Assume n is even. Then there is an integer k such that $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$, and since $2k^2$ is an integer, n^2 is even. $\square$

2.8.2 Activity. Let n be an integer. Prove each claim.

(a) n is odd if and only if $n + 2$ is odd.

(b) n is odd if and only if $n + 1$ is even.

There are also more complicated statements of equivalence:

2.8.3 Claim. *Let n be an integer. Then the following are equivalent:*

(i) n is even;

(ii) $n + 1$ is odd;

(iii) $n + 2$ is even.

A statement of the form **the following are equivalent**, which is often abbreviated “TFAE”, is followed by two or more statements (in our case, (i), (ii), and (iii)). When dealing with so many statements, writing “if...then” becomes cumbersome, so we often use the notation “ $\implies$ ” and the word “implies” instead. TFAE means that each of the following statements implies all of the others. In other words: (i) $\implies$ (ii) and (iii); (ii) $\implies$ (i) and (iii), and (iii) $\implies$ (i) and (ii). Another way of making the same claim is to say (i) $\iff$ (ii), (ii) $\iff$ (iii), and (iii) $\iff$ (i). Proving three “if and only if” statements would usually require six parts, but luckily we don’t have to do that much work. Instead, it suffices to prove a circular chain of implications involving all the statements. For us, this means proving (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (i) or proving (i) $\implies$ (iii) $\implies$ (ii) $\implies$ (i). If either one of these circular chains of implications has been established, then any one of the three statements implies both other statements. We’ll give a proof using the first chain of implications.

Proof. (i) $\implies$ (ii): Since n is even, there exists an integer k such that $n = 2k$. Then $n + 1 = 2k + 1$ is odd.

(ii) $\implies$ (iii): Since $n + 1$ is odd, there is an integer j such that $n + 1 = 2j + 1$. Then $n + 2 = 2j + 2 = 2(j + 1)$ is even.

(iii) $\implies$ (i): Since $n + 2$ is even, there is an integer l such that $n + 2 = 2l$. Then $n = 2l - 2 = 2(l - 1)$ is even. $\square$

That wasn’t so bad! We only had to prove three easy “if...then” statements. Here’s a similar claim for you to prove:

2.8.4 Activity. Let n be an integer. Prove that the following are equivalent:

- (i) n is odd;
- (ii) $n - 3$ is even;
- (iii) $n - 2$ is odd.

2.9 Practice

In 2.6 to 2.8, we let down our guard and trustingly tried to prove every claim we encountered. From now on, be skeptical and use the full four-step procedure in 2.3.2. Many of the claims below are false, and it is impossible to prove a false claim! If you think a claim is true, it is up to you to figure out whether to use a direct proof or a proof by contraposition or contradiction.

When turning in work proving or disproving a claim, don’t submit your scratch work. If a claim is false, your submitted solution should consist of “This claim is false!”, a counterexample, and a brief explanation of why it’s a counterexample.

2.9.1 Example.

- **Claim.** Let n be an integer. If n is odd, then $n^2 + 2n$ is even.

This claim is false! Let $n = 1$. Then n is odd since $1 = 2 \cdot 0 + 1$, but $n^2 + 2n$ is odd since $1^2 + 2 \cdot 1 = 3 = 2 \cdot 1 + 1$.

- **Claim.** Let m and n be integers. Then $m + 2n$ is odd if and only if $2m + n$ is odd.

This claim is false! Let $m = 0$ and $n = 1$. Then $m + 2n = 0 + 2 \cdot 1$ is even, but $2m + n = 2 \cdot 0 + 1$ is odd.

If a claim is true, write a precise, crisp proof. In most cases, your proof shouldn't be more than a few sentences.

2.9.2 Activity. Let m and n be integers. Prove or disprove each claim.

- (a) If m is even and n is odd, then mn is odd.
- (b) If mn is even, then m is even or n is even.
- (c) n is even if and only if $6n$ is even.
- (d) n is even if and only if $7n + 4$ is even.

2.9.3 Activity. Let r and s be real numbers. Prove or disprove each claim.

- (a) If r and s are rational, then $r + s$ and rs are rational.
- (b) If r is rational and s is irrational, then $r + s$ is irrational.
- (c) If r is rational and s is irrational, then rs is irrational.
- (d) r is rational if and only if r^2 is rational.

Chapter 3

Sets

Now that we have some experience writing proofs, we can begin a rigorous study of mathematics. A natural place to start is with one of the most basic and important mathematical objects: sets.

3.1 Definition of set

3.1.1 Definition. A **set** is an unordered collection of objects, called **elements** of the set. A set is said to **contain** its elements. If A is a set, we write $a \in A$ to denote that a is an element of A , and $a \notin A$ if a is not an element of A .

Two ways to describe a set are:

- (1) List all of its elements between a pair of curly braces ($\{$ and $\}$). If the elements follow an obvious pattern, then you can use an ellipsis ($\dots$) to indicate that the pattern continues.
- (2) Given a set A , you can create a new set B by defining properties that an element of the original set must satisfy to be in the new set. This is done by writing

$$B = \{ a \in A : \text{properties} \}.$$

The colon ($:$) means “such that” and can be replaced by a vertical line ($|$).

3.1.2 Example. Here are some sets.

- The set with no elements, called the **empty set** and denoted $\emptyset$ or $\{ \}$.
- $\{\text{kiwi, dragon, } \phi, \text{ turtle, krt, } 7\}$
- $\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$, the integers
- $\mathbb{Z}_{\geq 1} = \{ 1, 2, 3, \dots \} = \{ n \in \mathbb{Z} : n \geq 1 \}$, the positive integers
- For any $k \in \mathbb{Z}$, $\mathbb{Z}_{\geq k} = \{ n \in \mathbb{Z} : n \geq k \}$, the integers greater than or equal to k

- For any $d \in \mathbb{Z}$, $d\mathbb{Z} = \{n \in \mathbb{Z} : n = dk \text{ for some } k \in \mathbb{Z}\}$, the multiples of d
- $\mathbb{R}$, the real numbers
- For any $a, b \in \mathbb{R}$, $(a, b) = \{x \in \mathbb{R} : a < x < b\}$, an open interval of finite length
- For any $a \in \mathbb{R}$, $(a, \infty) = \{x \in \mathbb{R} : a < x\}$, an open interval of infinite length
- $\mathbb{Q} = \{x \in \mathbb{R} : x \text{ is rational}\}$, the rational numbers

For $a, b \in \mathbb{R}$, the half-open intervals $[a, b)$, $(a, b]$, and $[a, \infty)$ and the closed interval $[a, b]$ are defined in a manner similar to the open interval (a, b) by replacing the strict inequality $<$ by $\leq$ on the side where there is a square bracket. We can also modify the definition of (a, ∞) to define the open interval $(-\infty, b)$ and the half-open interval $(-\infty, b]$.

3.1.3 Note.

- Changing the order in which elements of a set are listed does not change the set. For instance, $\{1, 0\} = \{0, 1\}$. Listing an element repeatedly also does not change the set. For instance, $\{1, 1, 0, 1\} = \{0, 1\}$.
- A common convention is to use letters in the middle of the alphabet (such as k, m, n) to denote integers and letters near the end of the alphabet (such as x, y, z) to denote real numbers. Letters near the beginning of the alphabet (such as a, b, c) can be used in either context. In any case, we should always specify the meaning of the variables we introduce.
- Both $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}_{\geq 1}$ are sometimes referred to as the set of “natural numbers”. We will avoid ambiguity by using the terms “non-negative integers” and “positive integers”.
- In Chapters 1 and 2, we often started a claim with a sentence like “Let n be an integer.” We can now shorten such a sentence to “Let $n \in \mathbb{Z}$.”

3.1.4 Definition. Let A and B be sets.

- A is a **subset** of B , written $A \subseteq B$ or $B \supseteq A$, if every element of A is an element of B .
- A and B are **equal**, written $A = B$, if $A \subseteq B$ and $B \subseteq A$. If A and B are not equal, we write $A \neq B$.
- A is a **proper subset** of B , written $A \subsetneq B$, if A is a subset of B and $A \neq B$.

3.1.5 Example.

- The empty set is a subset of every set.
- $\mathbb{Z}_{\geq 1} \subsetneq \mathbb{Z}_{\geq 0} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}$ and $(0, 3) \subsetneq (0, \infty) \subsetneq \mathbb{R}$.

3.1.6 Note (Proving facts about subsets).

to prove that ____	show that ____
$A \subseteq B$	an arbitrary element of A must also be in B
$A \not\subseteq B$	there is an element of A that is not in B
$A = B$	$A \subseteq B$ and $B \subseteq A$
$A \neq B$	$A \not\subseteq B$ or $B \not\subseteq A$
$A \subsetneq B$	$A \subseteq B$ and $B \not\subseteq A$

3.1.7 Example.

- **Claim.** $4\mathbb{Z} \subseteq 2\mathbb{Z}$.

Proof. Let $n \in 4\mathbb{Z}$. Then there exists k such that $n = 4k$. Thus $n = 2(2k)$, so $n \in 2\mathbb{Z}$. $\square$

- **Claim.** $(2, 6) \subsetneq (2, 6]$.

Proof. Let $x \in (2, 6)$. Then $2 < x < 6$, so $2 < x \leq 6$. Thus $x \in (2, 6]$. To see that $(2, 6) \neq (2, 6]$, note that $6 \in (2, 6]$ but $6 \notin (2, 6)$. $\square$

3.2 Set operations

3.2.1 Definition. Let A and B be sets.

- The **union** of A and B is the set

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

- The **intersection** of A and B is the set

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

- A and B are called **disjoint** if $A \cap B = \emptyset$. If A and B are disjoint, we can refer to the union $A \cup B$ as a **disjoint union** and denote it by $A \sqcup B$.
- If it is clear from context that A is a subset of a set U , then the **complement** of A refers to the set

$$A^c = \{x \in U : x \notin A\}.$$

- The **set difference** of A and B is the set

$$A \setminus B = \{x \in A : x \notin B\}.$$

3.2.2 Example.

- Let $A = \{0, 1, 2\}$ and $B = \{2, 3\}$. Then $A \cup B = \{0, 1, 2, 3\}$, $A \cap B = \{2\}$, $A \setminus B = \{0, 1\}$, and $B \setminus A = \{3\}$. We can't write $A \sqcup B$ since A and B are not disjoint.
- Consider $[2, 7)^c$. The complement notation assumes that the half-open interval $[2, 7)$ is viewed as a subset of another set; the natural choice for this other set is $\mathbb{R}$. By definition, $x \in [2, 7)$ means that $2 \leq x < 7$, namely that $x \geq 2$ and $x < 7$. Thus $x \notin [2, 7)$ means that $x < 2$ or $x \geq 7$.

Claim. $[2, 7)^c = (-\infty, 2) \cup [7, \infty)$.

To prove two sets are equal, we show that each set is a subset of the other. In this case, we need to show $[2, 7)^c \subseteq (-\infty, 2) \cup [7, \infty)$ and $[2, 7)^c \supseteq (-\infty, 2) \cup [7, \infty)$. As writing out such statements is cumbersome, we can use the shorthand of labeling the two parts of the proof “ $\subseteq$ ” and “ $\supseteq$ ”.

Proof. $\subseteq$: Let $x \in [2, 7)^c$. Then $x \notin [2, 7)$, so $x < 2$ or $x \geq 7$. Thus $x \in (-\infty, 2)$ or $x \in [7, \infty)$.

$\supseteq$: Let $x \in (-\infty, 2) \cup [7, \infty)$. Then $x < 2$ or $x \geq 7$. Thus $x \notin [2, 7)$. □

Note that we could also write the union in the claim as a disjoint union.

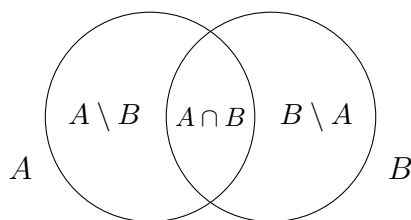
3.2.3 Activity. Prove or disprove each claim.

- (a) $5\mathbb{Z} \subseteq 10\mathbb{Z}$.
- (b) $3\mathbb{Z} \subseteq (2\mathbb{Z})^c$.
- (c) $(1, 3) \cup [3, 7] = (1, 7]$.
- (d) $(1, 3) \cap [3, 7] = \emptyset$.
- (e) $(-\infty, 3) \cap (-2, \infty) = \emptyset$.
- (f) $\mathbb{R} \setminus [-2, 4) = (-\infty, -2) \cup [4, \infty)$.

3.3 Set identities

In the previous section, we considered claims involving specific subsets of $\mathbb{Z}$ and $\mathbb{R}$. We now take a broader view, considering general statements that are fundamentally important as they apply to all sets.

Venn diagrams are useful tools for visualizing union, intersection, disjoint union, complement, and set difference of abstract sets. Visualize two general sets A and B as the interiors of two overlapping circles:



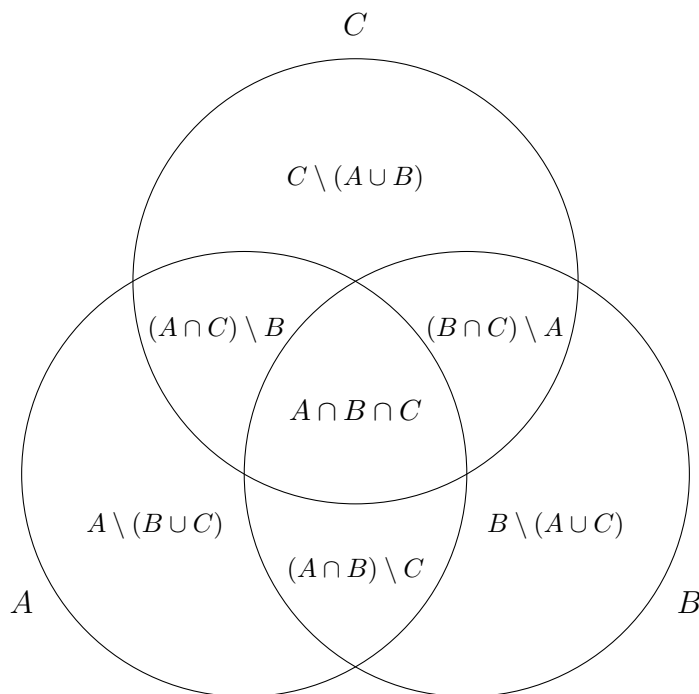
The labels “ A ” and “ B ” distinguish the two circles. The combined interiors of the two circles is $A \cup B$, which is not labeled. The labels $A \setminus B$, $A \cap B$, and $B \setminus A$ show how to break up $A \cup B$ into three smaller disjoint regions. As the diagram suggests, there are identities

- $(A \setminus B) \sqcup (A \cap B) = A$,
- $(A \cap B) \sqcup (B \setminus A) = B$, and
- $(A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A) = A \cup B$,

where we have used “ $\sqcup$ ” when the unions are disjoint. We will rigorously prove some of these identities shortly.

A Venn diagram is useful as a tool for distinguishing general intersections, unions, and differences. It should not be taken too literally in any specific context. For instance, the labeled regions inside the circles are disjoint, but it is unclear how many elements are in each labeled region. A could be empty, in which case the interior of the A -circle contains no elements. Although the circles for A and B overlap, A and B could still be disjoint, as $A \cap B$ could be empty.

For three sets, the Venn diagram looks like



Make sure you understand why the labels on each region are correct. The diagram suggests the identity

$$((A \cap B) \setminus C) \sqcup (A \cap B \cap C) = A \cap B$$

among many others. As in the case of two sets, the diagram makes no assumptions about how many elements are in each region.

3.3.1 Example. Let's find an identity for $A \cap (B \cup C)$. The Venn diagram above suggests that $A \cap (B \cup C)$ can be written as the disjoint union of the three sets $(A \cap C) \setminus B$, $A \cap B \cap C$, and $(A \cap B) \setminus C$. This would be a fairly complicated statement, so let's look for something simpler. The disjoint union of these three sets can be written as the union of the relatively simple sets $A \cap B$ and $A \cap C$.

Claim. Let A , B , and C be sets. Then $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof. $\subseteq$: If $a \in A \cap (B \cup C)$, then $a \in A$ and $a \in B \cup C$. Thus $a \in B$ or $a \in C$, so $a \in A \cap B$ or $a \in A \cap C$, namely $a \in (A \cap B) \cup (A \cap C)$.

$\supseteq$: If $a \in (A \cap B) \cup (A \cap C)$, then $a \in A \cap B$ or $a \in A \cap C$, namely $a \in A$ and $a \in B$, or $a \in A$ and $a \in C$. In either case $a \in A$, and $a \in B$ or $a \in C$, namely $a \in A$ and $a \in B \cup C$. Thus $a \in A \cap (B \cup C)$. $\square$

Proofs like these are technical, and it is easy to get confused. Still, it is important to be comfortable working with sets in this manner. Keeping a Venn diagram handy can help.

3.3.2 Note (Proving claims about abstract sets).

- Use the four-step procedure 2.3.2.
- A Venn diagram can help you decide whether the claim is likely to be true or false. If two descriptions of sets yield different regions of a Venn diagram, then create a simple counterexample for which one of the regions that is in one set but not the other is non-empty.
- When a claim involves the complement of an abstract set, you may assume there is some “universal set” U containing all sets mentioned in the claim. You can visualize U by drawing a rectangle around the Venn diagram and viewing its interior as U .

3.3.3 Example.

- **Claim.** $(A \cup B)^c = A^c \cup B^c$.

Using Venn diagrams, the set $(A \cup B)^c$ is inside U but outside $A \cup B$. The set $A^c \cup B^c$ looks larger, as it also contains $B \setminus A$ and $A \setminus B$. We construct a counterexample by defining A and B in such a way that $B \setminus A$ or $A \setminus B$ is non-empty. Here is our solution:

This claim is false! Let $A = \{a\}$ and $B = \emptyset$. Then $A \cup B = \{a\}$, so $a \notin (A \cup B)^c$. However, $a \notin B$, so $a \in B^c$, and thus $a \in A^c \cup B^c$.

3.3.4 Activity. Let A , B , and C be sets. Prove or disprove each claim.

- (a) $(A \cap B)^c = A^c \cup B^c$.
- (b) $(A^c)^c = A$.
- (c) $A \setminus B = A \cap B^c$.
- (d) $(A \cup B) \setminus B = A$.
- (e) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
- (f) $A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$.
- (g) If $A \cup C = B \cup C$, then $A = B$.
- (h) If $A \cup C = B \cup C$ and $A \cap C = B \cap C$, then $A = B$.

3.4 Advanced set constructions

Given two sets A and B , we can think of $A \cup B$ and $A \setminus B$ as the “sum” and “difference” of A and B . The following definition provides a notion of “product” of A and B .

3.4.1 Definition. Let A and B be sets. The (Cartesian) **product** of A and B is the set

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Elements of $A \times B$ are called **ordered pairs**.

3.4.2 Example.

- Let $A = \{1, 2, 3\}$ and $B = \{u, v\}$. Then

$$A \times B = \{(1, u), (1, v), (2, u), (2, v), (3, u), (3, v)\}.$$

- Let A be any set. Then $A \times \emptyset = \emptyset$.
- $\mathbb{R} \times \mathbb{R}$ is the familiar (x, y) -plane from calculus.

The product of A and B will be used in Chapters 6 and 7 to define the graph of a function and the graph of a relation.

We can extend the notion of union, intersection, and product to arbitrary (possibly infinite) collections of sets.

3.4.3 Definition. Given any set I and sets A_i for each $i \in I$, we define

$$\begin{aligned}\bigcup_{i \in I} A_i &= \{x : x \in A_i \text{ for some } i \in I\} \\ \bigcap_{i \in I} A_i &= \{x : x \in A_i \text{ for all } i \in I\} \\ \prod_{i \in I} A_i &= \{(x_i)_{i \in I} : x_i \in A_i \text{ for all } i \in I\}\end{aligned}$$

The set I is called an **index set** and the sets A_i are a **collection** of sets indexed by I . In the particular case $I = \{1, 2, 3, \dots, n\}$, we denote these sets by

$$\bigcup_{i=1}^n A_i, \quad \bigcap_{i=1}^n A_i, \quad \prod_{i=1}^n A_i.$$

In the case $I = \mathbb{Z}_{\geq 1}$, we use similar notation with n replaced by ∞ .

3.4.4 Example.

- $\bigcup_{i=1}^{\infty} (-i, i) = \mathbb{R}$ and $\bigcap_{i=1}^{\infty} (-i, i) = \{0\}$, where $(-i, i)$ denotes the open interval in $\mathbb{R}$.

- $\prod_{i=1}^n A$ is denoted A^n . The elements of A^n are called ***n*-tuples**, **lists**, or **sequences** in A . For instance, $\mathbb{R}^n$ contains *n*-tuples $(x_1, x_2, \dots, x_n)$ of real numbers. Note that the x_i in an *n*-tuple need not be distinct.
- $\prod_{i=1}^\infty A$ is denoted A^∞ . Its elements are called **infinite sequences** or **infinite lists** in A . For instance, $\mathbb{R}^\infty$ contains infinite sequences $(x_1, x_2, x_3, \dots)$. We will study limits of such sequences in Chapter 8.

3.4.5 Definition. Given a set A , the **power set** of A , denoted $\mathcal{P}(A)$, is the set containing all subsets of A .

Note that every *element* of the set $\mathcal{P}(A)$ is a *set*.

3.4.6 Example.

- $\mathcal{P}(\emptyset) = \{\emptyset\}$
- $\mathcal{P}(\{\alpha\}) = \{\emptyset, \{\alpha\}\}$
- $\mathcal{P}(\{0, 1\}) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$
- $\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$
- For every $d \in \mathbb{Z}$, the set $d\mathbb{Z}$ is an element of $\mathcal{P}(\mathbb{Z})$
- Every open/half-open/closed interval is an element of $\mathcal{P}(\mathbb{R})$

3.5 Cardinality

3.5.1 Definition. A set A is **finite** if it contains finitely many elements. When A is finite, the **cardinality** of A , denoted $|A|$, is the number of elements of A .

Note that $|A| \in \mathbb{Z}_{\geq 0}$; in particular, the cardinality of a set is a *number*, not a set. Cardinality can be interpreted as a measure of the “size” of a set.

3.5.2 Example.

- $|\emptyset| = 0$
- $|\{17, \text{apple}, 17, 17\}| = 2$
- $|\{1, 3, 5, 7, \dots, 19\}| = 10$

When proving claims about cardinality of abstract sets, a useful technique is to choose a *labeling* of the elements of the sets. For instance, if A is a finite set, let $n = |A|$. Then we can write $A = \{a_1, a_2, \dots, a_n\}$, where $a_1, \dots, a_n$ are variables representing the elements of A . Moreover, in the context of another finite set B with $m = |B|$, we can write

$$A = \{c_1, \dots, c_k, a_{k+1}, \dots, a_n\} \quad \text{and} \quad B = \{c_1, \dots, c_k, b_{k+1}, \dots, b_m\},$$

where $\{c_1, \dots, c_k\}$ is a labeling of $A \cap B$ with $k = |A \cap B|$. These refined labelings allow us to describe other sets such as $A \setminus B$, $B \setminus A$, and $A \cup B$ precisely enough to calculate their cardinality in terms of n, m, k , as in the following activity.

3.5.3 Activity. Let A and B be finite sets. Prove or disprove the following claims. For all claims that are false, adjust the right side to make the claim true and prove the new claim.

(a) $|A \cup B| = |A| + |B|$.

(b) $|A \setminus B| = |A| - |B|$.

(c) $|A \times B| = |A| \cdot |B|$.

Cardinalities can also be defined for infinite sets in such a way that the “sizes” of different infinite sets can be compared. One of the first big questions related to the cardinality of an infinite set A is whether it is possible to put the elements of A into an infinite list.

3.5.4 Definition. A set A is **countable** if it is finite or if its elements can be put into an infinite list; in the latter case it is **countably infinite**. A set that is not countable is **uncountable**.

3.5.5 Example.

- Every finite set is countable.
- $\mathbb{Z}_{\geq 1}$ is countable (more precisely, countably infinite) since its elements can be listed as $(1, 2, 3, 4, 5, \dots)$.
- The set of positive even integers is countable since its elements can be listed as $(2, 4, 6, \dots)$.
- $\mathbb{Z}$ is countable since its elements can be listed as $(0, 1, -1, 2, -2, 3, -3, \dots)$.

As lists are ordered while sets are unordered, we write lists using parentheses rather than curly braces.

Are the rational numbers countable? Two observations:

- There are infinitely many fractions with denominator 2, infinitely many fractions with denominator 3, and so on; namely, an infinite number of denominators and an infinite number of fractions for each denominator.
- The rational numbers have the property that if $x, y \in \mathbb{Q}$ with $x < y$, then there are infinitely many other rational numbers between x and y .

With these points in mind, the prospect of putting the rational numbers into a single list sounds absurd. However, recall from the example of $\mathbb{Z}$ that the listing of the elements of a set of numbers does not have to reflect the natural ordering of the set. When dealing with infinity, the seemingly absurd is often possible.

3.5.6 Proposition. *Let $A_1, A_2, A_3, \dots$ be a countable collection of sets, each of which is countable. Then $\bigcup_{i=1}^{\infty} A_i$ is countable.*

Recall that “countable” includes the possibility of “finite”. We prove the proposition in the most difficult case, when the collection is countably infinite and each A_i is countably infinite.

Proof. Since A_i is countable, its elements can be listed as $(a_{i1}, a_{i2}, a_{i3}, \dots)$. The elements of $\bigcup_{i=1}^{\infty} A_i$ can be viewed in an infinite array in which the i th row contains the elements of A_i :

$$\begin{array}{ccccccc} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & \cdots & & \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & \cdots & & \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & \cdots & & \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & \cdots & & \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \end{array}$$

Consider the upward-sloping diagonals of the array, each of which contains finitely many elements. The first diagonal contains $a_{1,1}$, the second diagonal contains $a_{2,1}$ and $a_{1,2}$, the third contains $a_{3,1}$, $a_{2,2}$, and $a_{1,3}$, and so on. Combining the lists of the elements of each diagonal yields

$$\bigcup_{i=1}^{\infty} A_i = (a_{1,1}, a_{2,1}, a_{1,2}, a_{3,1}, a_{2,2}, a_{1,3}, a_{4,1}, a_{3,2}, a_{2,3}, a_{1,4}, a_{5,1}, \dots). \quad \square$$

3.5.7 Corollary. $\mathbb{Q}$ is countable.

Proof. For each $i \in \mathbb{Z}_{\geq 1}$, let A_i denote the set of all fractions with denominator i . The elements of A_i can be listed as

$$\left(\frac{0}{i}, \frac{1}{i}, \frac{-1}{i}, \frac{2}{i}, \frac{-2}{i}, \frac{3}{i}, \frac{-3}{i}, \dots \right),$$

so A_i is countable. Since $\bigcup_{i=1}^{\infty} A_i = \mathbb{Q}$, 3.5.6 implies that $\mathbb{Q}$ is countable. $\square$

Can we push these methods even further to show that $\mathbb{R}$ is countable? The following famous argument shows that no amount of cleverness will allow us to put the real numbers into a list. The proof uses an interpretation of real numbers as infinite decimals and features a technique called “diagonalization”.

3.5.8 Theorem. $\mathbb{R}$ is uncountable.

Proof. It suffices to prove that the open interval $(0, 1)$ is uncountable. Assume for contradiction that the elements of $(0, 1)$ can be listed as $(x_1, x_2, x_3, \dots)$. Each x_i can be written as an infinite decimal $x_i = 0.a_{i,1}a_{i,2}a_{i,3}\dots$, where $a_{i,j}$ are the decimal digits of x_i . (Recall that decimal digits are integers between 0 and 9.) Consider the array of all decimal digits of all x_i :

$$\begin{array}{ccccccc}
a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & \cdots & & \\
a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & \cdots & & \\
a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & \cdots & & \\
a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & \cdots & & \\
\vdots & \vdots & \vdots & \vdots & \ddots & &
\end{array}$$

Define a real number y as an infinite decimal $y = 0.b_1b_2b_3\dots$ by choosing each b_i to be any integer between 0 and 9 that is not equal to $a_{i,i}$. Since the i th digits of y and x_i are different, $y \neq x_i$. Thus y does not appear on the list $(x_1, x_2, x_3, \dots)$, which contradicts the assumption that all elements of $(0, 1)$ can be listed. $\square$

For a full definition of cardinality, see Problem 17.

3.6 Problems

Problem 2. Let A be a finite set. State and prove a formula for $|\mathcal{P}(A)|$ in terms of $|A|$. (*Hint: consider the examples in 3.4.6.*)

Problem 3. Let $\mathcal{S}$ denote the set containing all sets, namely

$$\mathcal{S} = \{ A : A \text{ is a set} \}.$$

Consider the subset $\mathcal{R} \subseteq \mathcal{S}$ containing only those sets that do not contain themselves:

$$\mathcal{R} = \{ A \in \mathcal{S} : A \notin A \}.$$

- (a) Explain why $\mathcal{S}$ and $\mathcal{R}$ are elements of $\mathcal{S}$.
- (b) Is $\mathcal{S}$ an element of $\mathcal{R}$? Explain.
- (c) Is $\mathcal{R}$ an element of $\mathcal{R}$? Explain.

Due to the issues observed in this problem, additional set-theoretic nuances must be discussed to rigorously define $\mathcal{S}$, the “set” containing all sets. Since such a discussion is beyond the scope of this text, we’ll settle for using quotation marks around the word “set” when describing $\mathcal{S}$ in later chapters as a reminder of these nuances.

Chapter 4

Number Theory

We now begin a study of number theory, a branch of mathematics that is primarily concerned with the properties of the integers, $\mathbb{Z}$.

4.1 Divisibility

4.1.1 Definition. If $a, b \in \mathbb{Z}$, then a **divides** b (or b is **divisible** by a), written $a \mid b$, if there is an integer k such that $b = ak$. We then call a a **factor** or **divisor** of b and call b a **multiple** of a . We write $a \nmid b$ if a does not divide b .

Note that “ $a \mid b$ ” is a statement expressing a relationship that a and b satisfy, which can be true or false depending on a and b . This is different than a/b or b/a , which denote rational numbers obtained by dividing two integers. For instance, a statement like “ $2 \mid 6 = 3$ ” does not make sense. Think of $a \mid b$ as a statement about divisibility, while a/b is division. Since division requires the introduction of rational numbers, divisibility is the more basic notion in number theory.

4.1.2 Example.

- 6 divides 0, 6, -6 , 12, and -12 because these integers can be written as $6 \cdot 0$, $6 \cdot 1$, $6 \cdot (-1)$, $6 \cdot 2$, and $6 \cdot (-2)$.
- Let $n \in \mathbb{Z}$. Then $2 \mid n$ if and only if n is even.
- $1 \mid a$ for every $a \in \mathbb{Z}$.
- $a \mid 0$ for every nonzero $a \in \mathbb{Z}$.
- $0 \mid a$ if and only if $a = 0$.

As mentioned in the example, an integer being even means that the integer is divisible by 2. Divisibility generalizes the property of being “even”. The important properties of divisibility in the following proposition can be proved in a manner similar to statements about even integers in Chapter 2.

4.1.3 Proposition. Let $a, b, c \in \mathbb{Z}$. Then

- (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
- (ii) if $a \mid b$, then $a \mid bc$;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

4.1.4 Activity.

- (a) Prove the proposition.
- (b) Prove that if $a, b, c \in \mathbb{Z}$ satisfy $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any $m, n \in \mathbb{Z}$.

4.1.5 Activity. Let $a, b, c \in \mathbb{Z}$. Prove or disprove each claim.

- (a) If $a \nmid b$, then $a \nmid bc$.
- (b) If $a \mid bc$, then $a \mid b$ or $a \mid c$.
- (c) If $ab \mid ac$ and $a \neq 0$, then $b \mid c$.

4.2 Prime numbers

Let's get reacquainted with our old friends from elementary school, the prime numbers.

4.2.1 Definition. An integer $p \geq 2$ is **prime** if its only positive factors are 1 and p . An integer ≥ 2 is called **composite** if it is not prime.

Note that 1 is not prime. We'll see shortly why we want to exclude 1.

4.2.2 Activity. We can use the *sieve of Eratosthenes* to compute all the primes less than a given number. For instance, to find all primes $\leq N$:

- *Step 1:* Write the integers from 2 to N in ascending order.
- *Step 2:* Circle the smallest unmarked number (it is prime) and cross out all other multiples of that number (they are composite).
- *Step 3:* Repeat Step 2 until all numbers are circled or crossed out. The circled numbers are the primes!

Use the sieve of Eratosthenes to find all primes ≤ 100 .

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

How many primes are there in total? The following famous argument provides an answer.

4.2.3 Theorem. *There are infinitely many primes.*

Proof. Assume for contradiction that there are only finitely many primes, say $\{p_1, p_2, \dots, p_n\}$. Then consider

$$N = p_1 p_2 \cdots p_n + 1.$$

Since $N \geq 2$, N has a prime factorization. Thus there is a p_i that divides N . But p_i also divides $p_1 p_2 \cdots p_n$, so p_i divides $N - p_1 p_2 \cdots p_n = 1$ by 4.1.4(b). This is a contradiction since no prime can divide 1. $\square$

Primes are important because they are the multiplicative building blocks of the integers.

4.2.4 Definition. Let $n \in \mathbb{Z}_{\geq 2}$. A **prime factorization** of n is an equation

$$n = p_1 p_2 \cdots p_r$$

in which $r \in \mathbb{Z}_{\geq 1}$ is the number of factors and $p_1, \dots, p_r$ are primes.

4.2.5 Example (Examples of prime factorizations).

- $6 = 2 \cdot 3 = 3 \cdot 2$
- $13 = 13$
- $63 = 3 \cdot 3 \cdot 7 = 3 \cdot 7 \cdot 3 = 7 \cdot 3 \cdot 3$
- $100 = 2 \cdot 2 \cdot 5 \cdot 5$

The key theorem regarding primes is that every integer ≥ 2 has a prime factorization and that the primes in the factorization are unique up to reordering.

4.2.6 Theorem (Fundamental theorem of arithmetic). *Let $n \in \mathbb{Z}_{\geq 2}$. Then*

- (a) *n has a prime factorization $n = p_1 p_2 \cdots p_r$;*
- (b) *The list of primes $p_1, \dots, p_r$ is unique up to reordering.*

4.2.7 Note.

- An integer is prime if and only if the prime factorization contains only one factor (itself). Thus an integer is composite if and only if its prime factorization has at least two factors (possibly the same prime twice).
- If 1 were prime, then the prime factorization would not be unique since any number of 1's could be included as factors.

The idea of the proof of 4.2.6 is that we can find the prime factorization of a composite number n by first finding a divisor d of n , obtaining an equation $n = de$ for some integer e , and then continuing this process for d and e until all the factors are prime. Making this recursive argument rigorous requires proof by strong induction. The full proof is in 5.2.

The fundamental theorem of arithmetic can be used to deduce important divisibility properties related to primes.

4.2.8 Claim. *Let $n \in \mathbb{Z}_{\geq 2}$ be an integer and $p \in \mathbb{Z}_{\geq 2}$ be prime. Then $p \mid n$ if and only if p is one of the prime factors in the prime factorization of n .*

Proof. $\implies$: Since $p \mid n$, $n = pk$ for some $k \in \mathbb{Z}$. If $k = 1$, then $n = p$ is the prime factorization of n . If $k \geq 2$, let $k = q_1 q_2 \cdots q_s$ denote the prime factorization of k . Then the prime factorization of n is $n = p q_1 q_2 \cdots q_s$. Thus p is one of the primes in the prime factorization of n .

$\impliedby$: Since p is one of the primes in the prime factorization of n , we can write the prime factorization as $n = p p_2 \cdots p_r = p(p_2 \cdots p_r)$. Thus $p \mid n$. $\square$

4.2.9 Proposition. *Let $a, b \in \mathbb{Z}$ and $p \in \mathbb{Z}_{\geq 2}$ be prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof by contraposition. Assume that $p \nmid a$ and $p \nmid b$. Let the prime factorizations of a and b be $a = p_1 \cdots p_r$ and $b = q_1 \cdots q_s$. Since $p \nmid a$ and $p \nmid b$, p does not appear in either of these factorizations. Thus the prime factorization of ab , which is $ab = p_1 \cdots p_r q_1 \cdots q_s$, does not include p as a factor. So $p \nmid ab$. $\square$

As written, the proof of 4.2.9 requires $a, b \geq 2$ in order to make use of prime factorizations. The remaining cases are not difficult to check. We can easily reduce to the case where a, b are both non-negative, as $p \mid a$ if and only if $p \mid -a$. The cases where a or b is equal to 0 or 1 can be checked individually.

The statement in 4.2.9 is a special feature of prime numbers that fails if p is composite. See 4.1.5.

4.3 Modular arithmetic

In some situations we care mostly about remainders. For instance, what time will it be 50 hours from now? One way to find the answer is to compute the remainder of 50 divided by 24, and add that to the current time. Working with remainders is surprisingly interesting mathematically.

Recall division with remainder from elementary school. For instance, when dividing 23 by 4, we note that 4 goes into 23 five times, leaving a remainder of 3. Expressing this result as an equation, we write

$$23 = 4 \cdot 5 + 3.$$

This is the most basic notion of “division” in number theory, as it involves only integers. The general statement is as follows.

4.3.1 Theorem (Division with remainder). *Let $a, m \in \mathbb{Z}$ with $m \geq 1$. Then there are unique integers q, r with $0 \leq r < m$ such that $a = mq + r$.*

The idea of the proof is to start with a and keep adding or subtracting m until the result is in the set $\{0, 1, \dots, m-1\}$. This can be made rigorous using induction; see Problem 10.

4.3.2 Definition. In the equality $a = mq + r$ in the theorem, m is the **divisor**, a is the **dividend**, q is the **quotient**, and r is the **remainder**. To specify where the remainder came from, we say that r is the remainder of a **modulo** m , denoted $r = a \bmod m$.

The word “modulo” has the same meaning as “when divided by”.

4.3.3 Example. Applying the division algorithm to $a = -7$ and $d = 6$, we get $q = -2$ and $r = 5$, namely $-7 = -2 \cdot 6 + 5$. So $-7 \bmod 6 = 5$.

4.3.4 Definition. If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_{\geq 1}$, then a is **congruent to b modulo m** , written $a \equiv_m b$, if $a \bmod m = b \bmod m$. We say that $a \equiv_m b$ is a **congruence** (rather than an *equation*) and that m is its **modulus**. If a and b are not congruent modulo m , we write $a \not\equiv_m b$.

4.3.5 Example. $11 \equiv_4 7 \equiv_4 3 \equiv_4 -1 \equiv_4 -5$ because all these numbers have remainder 3 when divided by 4. For example, $-1 = -1 \cdot 4 + 3$.

The following result provides two statements that are equivalent to our original definition of congruence modulo m and don’t mention remainders.

4.3.6 Proposition. *Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_{\geq 1}$. Then the following are equivalent:*

- (i) $a \equiv_m b$;
- (ii) $a = b + mk$ for some $k \in \mathbb{Z}$;
- (iii) $m \mid (a - b)$.

Proof. (i) $\implies$ (ii): Suppose $a \equiv_m b$, namely that $a \bmod m = r = b \bmod m$. Thus there are integers q_1 and q_2 such that $a = mq_1 + r$ and $b = mq_2 + r$, so $a - mq_1 = r = b - mq_2$. Adding mq_1 to both sides, we see that $a = b + m(q_1 - q_2)$.

(ii) $\implies$ (iii): Suppose $a = b + mk$ for some $k \in \mathbb{Z}$. Then $a - b = mk$, so $m \mid (a - b)$.

(iii) $\implies$ (i): Suppose $m \mid (a - b)$, namely there is a $k \in \mathbb{Z}$ such that $a - b = mk$. Applying division with remainder on a and m and again on b and m yields equations $a = mq_1 + r_1$ and $b = mq_2 + r_2$. We need to show that $r_1 = r_2$. Note that

$$mk = a - b = m(q_1 - q_2) + (r_1 - r_2),$$

which implies $r_1 - r_2 = m(k - q_1 + q_2)$. Thus m divides $r_1 - r_2$. But r_1 and r_2 are both between 0 and $m - 1$, so $-(m - 1) \leq r_1 - r_2 \leq m - 1$. The only multiple of m in this interval is 0, so we must have $r_1 - r_2 = 0$, namely $r_1 = r_2$. $\square$

Part (ii) in particular is more useful than our definition of $a \equiv_m b$, so use it whenever you need to prove something about congruences. Another way to think about (ii) is that adding a multiple of m to a number doesn't change its remainder when divided by m .

4.3.7 Example.

- Using (ii) of the proposition, we can easily see that $11 \equiv_4 7$ by noting that $11 = 7 + 4$.
- **Claim.** Let $a \in \mathbb{Z}$. If $a^2 \equiv_m 0$, then $a \equiv_m 0$.

This claim is false! Let $m = 4$ and $a = 2$. Then $a^2 = 4 \equiv_4 0$, but $a = 2 \not\equiv_4 0$.

4.3.8 Activity. Let $a \in \mathbb{Z}$. Prove or disprove each claim.

- If $a \equiv_m 1$, then $a^2 \equiv_m 1$.
- If $a \equiv_m 1$, then $a \equiv_{2m} 2$.
- If $a \equiv_{2m} 0$, then $a \equiv_m 0$.

A useful result that makes computations modulo m easy is the following:

4.3.9 Proposition (Substitution). *Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_{\geq 1}$. If $a \equiv_m b$ and c is any integer, then*

$$a + c \equiv_m b + c \quad \text{and} \quad ac \equiv_m bc.$$

4.3.10 Activity. Prove the proposition using 4.3.6(ii) as an equivalent definition of congruence modulo m .

4.3.9 says that working modulo m , we can replace any number in a sum or product by any other number with the same remainder, such as a number between 0 and $m - 1$ or sometimes a small negative number. For instance, to find the remainder of an expression a when divided by m , you could evaluate a and then divide by m in the elementary school sense 4.3.1 to find the remainder. But sometimes a is hard to evaluate because it may be hundreds of digits long! A better solution is to repeatedly use 4.3.9 to simplify the expression until you get a result between 0 and $m - 1$.

4.3.11 Example.

- Calculate $(3 \cdot 10^2 + 15) \bmod 7$. In elementary school, you would compute $3 \cdot 10^2 + 15 = 315$, and then use long division to see that $315 = 45 \cdot 7$, so the remainder is 0. Using congruences, we note that $10 \equiv_7 3$, $15 \equiv_7 1$, and $9 \equiv_7 2$, so

$$3 \cdot 10^2 + 15 \equiv_7 3 \cdot 3^2 + 1 = 3 \cdot 9 + 1 \equiv_7 3 \cdot 2 + 1 = 6 + 1 \equiv_7 0.$$

Thus $(3 \cdot 10^2 + 15) \bmod 7 = 0$.

- Calculate $12^{100} \bmod 11$. Now the elementary school method is much less appealing, but this is easy using congruences. Since $12 \equiv 1 \pmod{11}$,

$$12^{100} \equiv_{11} 1^{100} = 1,$$

so $12^{100} \bmod 11 = 1$.

- Calculate $10^{2013} \bmod 11$. Since $10 \equiv -1 \pmod{11}$,

$$10^{2013} \equiv_{11} (-1)^{2013} = -1 \equiv_{11} 10,$$

so $10^{2013} \bmod 11 = 10$.

- Calculate $3^{100} \bmod 7$. There's no trick quite as easy as in the previous examples, but the small modulus still makes things easy. Start by finding a power of 3 that is particularly easy to work with modulo 7: note that $3^3 = 27 \equiv_7 -1$. Thus

$$3^{100} = 3 \cdot (3^3)^{33} \equiv_7 3 \cdot (-1)^{33} = -3 \equiv_7 4,$$

so $3^{100} \bmod 7 = 4$.

In computations, we use the notation " $\equiv_m$ " when applying congruence and the usual " $=$ " when simplifying in the usual sense without using congruence. This makes the computation easier to follow because it pinpoints exactly where congruence is being used. For example, we wrote $3 \cdot 2 + 1 = 6 + 1 \equiv_7 0$. Since two integers that are equal are *always* congruent for any modulus, it would be correct but less precise to write the equality as $3 \cdot 2 + 1 \equiv_7 6 + 1$.

To summarize, arithmetic modulo m works just like normal arithmetic, except that you can replace numbers being added or multiplied (but not exponents) by congruent numbers modulo m . This makes modular arithmetic much *easier*, because you can keep the numbers smaller than the modulus.

4.3.12 Activity. Find the remainder.

- (a) $(3 \cdot 4 \cdot 5 + 11) \bmod 6$
- (b) $(-2^3 \cdot 11 - 71) \bmod 7$
- (c) $(5^{999} + 16) \bmod 4$

(d) $3^{2200} \bmod 8$

In addition to facilitating calculations involving large exponents, 4.3.9 implies that when proving a congruence modulo m containing an expression involving a variable a , it suffices to consider values of a in the set $\{0, 1, 2, \dots, m-1\}$. When m is small, checking the claim in each of these cases may be feasible.

4.3.13 Example.

- **Claim.** Let $a \in \mathbb{Z}$. Then $a^2 \not\equiv_7 6$.

Proof.

If $a \equiv_7 0$, then $a^2 \equiv_7 0$.

If $a \equiv_7 4$, then $a^2 \equiv_7 16 \equiv_7 2$.

If $a \equiv_7 1$, then $a^2 \equiv_7 1$.

If $a \equiv_7 5$, then $a^2 \equiv_7 25 \equiv_7 4$.

If $a \equiv_7 2$, then $a^2 \equiv_7 4$.

If $a \equiv_7 6$, then $a^2 \equiv_7 36 \equiv_7 1$.

If $a \equiv_7 3$, then $a^2 \equiv_7 9 \equiv_7 2$.

In all cases, $a^2 \not\equiv_7 6$. □

The technique in the example, which is similar to exhaustive proof, is not very insightful. Use it only if the modulus is small and if you don't see another path forward.

4.3.14 Activity. Let $a \in \mathbb{Z}$. Prove or disprove each claim.

(a) $a^2 \equiv_4 0$ or $a^2 \equiv_4 1$.

(b) If $a^2 \equiv_7 2$, then $a^{90} \equiv_7 1$.

(c) $-a \equiv_6 a$ if and only if $a \equiv_6 0$.

(d) $a \equiv_9 3$ if and only if $5a \equiv_9 6$.

Above, we often resorted to tricks to compute large powers of integers modulo m . The following theorem is a powerful tool for computing powers when the modulus m is prime:

4.3.15 Theorem (Fermat's little theorem). *If p is prime and $a \in \mathbb{Z}$ such that $p \nmid a$, then*

$$a^{p-1} \equiv_p 1.$$

The theorem says that for any a not divisible by p , p divides $a^{p-1} - 1$. For instance, if $3 \nmid a$, then $3 \mid (a^2 - 1)$. Indeed, 3 divides $1^2 - 1 = 0$, $2^2 - 1 = 3$, $4^2 - 1 = 15$, $5^2 - 1 = 24$, and so on.

For the proof of this theorem, see Problem 6.

4.3.16 Example. Calculate $3^{100} \bmod 7$. Since $7 \nmid 3$, we can apply Fermat's little theorem:

$$3^{100} = 3^4 \cdot (3^6)^{16} \equiv_7 3^4 \cdot (1)^{16} = 3^4 = 9^2 \equiv_7 2^2 = 4.$$

Note that this agrees with our answer in 4.3.11.

4.3.17 Activity. Use Fermat's little theorem to calculate each remainder.

(a) $7^{121} \bmod 13$

(b) $22^{100} \bmod 11$

(c) $24^{39} \bmod 7$

Congruences can be used to prove criteria for divisibility that involve the digits of a positive integer.

4.3.18 Claim. Let $a \in \mathbb{Z}_{\geq 1}$ and let D be the sum of the digits of a . Then $3 \mid a$ if and only if $3 \mid D$.

Proof. Write $a = a_n a_{n-1} \dots a_0$, where the a_i denote the digits of a . Then since $10 \equiv_3 1$,

$$a = a_0 + 10a_1 + 10^2a_2 + \dots + 10^na_n \equiv_3 a_0 + a_1 + \dots + a_n = D.$$

Thus $a \equiv_3 0$ if and only if $D \equiv_3 0$. □

4.4 Problems

Problem 4. Let $a, b \in \mathbb{Z}_{\geq 1}$. The **greatest common divisor** of a and b , denoted $\gcd(a, b)$, is the largest integer d such that $d \mid a$ and $d \mid b$. The **least common multiple** of a and b , denoted $\text{lcm}(a, b)$, is the smallest positive integer m such that $a \mid m$ and $b \mid m$.

(a) Determine the prime factorizations of 36 and 60. Using these prime factorizations, explain how to compute $\gcd(36, 60)$ and $\text{lcm}(36, 60)$.

(b) Let $a, b \in \mathbb{Z}_{\geq 1}$. Explain why $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$.

(Hint: if $a = 1$ or $b = 1$, it is easy to calculate $\gcd(a, b)$ and $\text{lcm}(a, b)$. If $a, b \geq 2$, use the prime factorizations of a, b .)

Problem 5. Let $m \in \mathbb{Z}_{\geq 1}$ and $a \in \mathbb{Z}$. We say that b is a **multiplicative inverse** of a modulo m if $ab \equiv_m 1$.

(a) Find a multiplicative inverse of 3 modulo 8.

(b) Find a multiplicative inverse of 4 modulo 8 or explain why no inverse exists.

For the rest of the problem, let p be prime and a be an integer not divisible by p .

- (c) Prove that no two of the integers $0, a, 2a, 3a, \dots, (p-1)a$ are congruent modulo p . (*Hint: use 4.2.9*).
- (d) Using (c), deduce that a has a multiplicative inverse modulo p . (*Hint: there are p numbers in the list in (c). What can be said about their remainders modulo p ?*)

Problem 6. Prove Fermat's little theorem (4.3.15) by completing the following outline:

Step 1: Let n be the product $1 \cdot 2 \cdots (p-1)$. Use Problem 5(c) to prove that $n \equiv_p a^{p-1}n$.

Step 2: Prove that n is not divisible by p . Therefore n has a multiplicative inverse modulo p by Problem 5(d).

Step 3: Use Steps 1 and 2 to deduce that $1 \equiv_p a^{p-1}$.

Problem 7. Let $a \in \mathbb{Z}_{\geq 1}$. Come up with a criterion involving the digits of a for when a is divisible by 9 and a criterion for when a is divisible by 11. Prove your criteria are correct.

Chapter 5

Induction

5.1 Proof by induction

Have you ever set up a domino rally? You place dominoes upright in a line, then knock over the first domino, which knocks over the second, which knocks over the third, and so on. Keep this image in mind while learning about mathematical induction, because the basic idea of induction is that of the domino rally.

Mathematical induction is a powerful technique for proving a claim is true for all positive integers. Here is a typical proof by induction:

5.1.1 Claim. *Let $n \in \mathbb{Z}_{\geq 1}$. Then the sum of the first n positive integers is*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Induction on n . The base case $n = 1$ is true since $1 = \frac{1(1+1)}{2}$. Now suppose the claim is true for some $n = k \in \mathbb{Z}_{\geq 1}$, namely that

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Then we compute

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k^2 + k + 2k + 2}{2} = \frac{(k+1)(k+2)}{2},$$

so the claim is true for $n = k + 1$. Thus all cases of the claim are true by induction. $\square$

Why did the previous argument prove the claim? Consider the statement “ $1 + 2 + \cdots + n = n(n+1)/2$ ”, which we’ll denote $S(n)$. The claim is stating that $S(n)$ is true for all $n \in \mathbb{Z}_{\geq 1}$. The proof above uses the following two-step approach.

5.1.2 Note (Proof by induction). Prove a claim $S(n)$ is true for all $n \in \mathbb{Z}_{\geq b}$ by:

(1) “Base case”: Prove $S(b)$ is true. (*Knock over the first domino.*)

- (2) “Inductive step”: Prove that $S(k)$ implies $S(k+1)$ for arbitrary $k \in \mathbb{Z}_{\geq b}$. (*Show that a falling domino knocks over the next domino.*)

For the claim we are considering, $b = 1$. These two steps prove the claim because (1) establishes $S(1)$, while (2) establishes the chain of implications

$$S(1) \implies S(2) \implies S(3) \implies S(4) \implies \dots$$

Since $S(1)$ is true, so is $S(2)$. Since $S(2)$ is true, so is $S(3)$. And so on. Each falling domino knocks over the next domino in line, so all of the infinitely many dominoes get knocked over.

5.1.3 Note.

- The power of induction is that for the inductive step, we get to assume a case $S(k)$, and use that to prove $S(k+1)$. For instance, our assumption that the claim is true for $n = k$ gave us a formula for adding up all the terms of $1 + 2 + \dots + k + (k+1)$ except the last one. This is often much easier than directly proving $S(k+1)$ without any assumptions.
- We do not use the $S(n)$ notation when writing a proof. Instead of writing $S(k)$, for instance, we refer to the theorem being true for $n = k$, as in the proof of 5.1.1.

Let’s try another proof by induction.

5.1.4 Claim. *Let $n \in \mathbb{Z}_{\geq 1}$. Then the sum of the first n odd positive integers is n^2 , namely*

$$1 + 3 + 5 + \dots + (2n - 3) + (2n - 1) = n^2.$$

Before starting a proof by induction, it is a good idea to check the first three cases to make sure the claim is plausible and then sketch a proof of the inductive step on scratch paper. For this claim, the first three cases are

$$1 = 1^2, \quad 1 + 3 = 2^2, \quad 1 + 3 + 5 = 3^2,$$

which are all true. For the inductive step, write the assumption $S(k)$ and the goal $S(k+1)$ on scratch paper and try to work out how to deduce the goal from the assumption. In this case,

The assumption $S(k)$ is that $1 + 3 + \dots + (2k - 3) + (2k - 1) = k^2$.

The goal $S(k+1)$ is that $1 + 3 + \dots + (2k - 3) + (2k - 1) + (2k + 1) = (k+1)^2$.

Note that the left side of the formula $S(k+1)$ equals the left side of the formula $S(k)$ plus $2k + 1$. Thus, using the assumption, the left side of $S(k+1)$ can be calculated as $k^2 + (2k + 1)$, which does equal $(k+1)^2$. We are now ready to write the proof:

Proof. Induction on n . The base case $n = 1$ is true since $1 = 1^2$. Now suppose the claim is true for some $n = k \in \mathbb{Z}_{\geq 1}$, namely that

$$1 + 3 + \cdots + (2k - 3) + (2k - 1) = k^2.$$

We can use this to compute

$$1 + 3 + \cdots + (2k - 3) + (2k - 1) + (2k + 1) = k^2 + (2k + 1) = (k + 1)^2,$$

which shows that the claim is true for $n = k + 1$. Thus all cases of the claim are true by induction. $\square$

5.1.5 Activity. Let $n \in \mathbb{Z}_{\geq 1}$. Prove each claim by induction or disprove by giving a counterexample.

(a) The sum of the first n even positive integers is $n(n + 1)$, namely $2 + 4 + \cdots + 2n = n(n + 1)$.

(b) $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n + 1) = \frac{n(n + 1)(n + 2)}{3}$.

(c) $1^2 + 2^2 + \cdots + n^2 = \frac{n(n + 1)(2n + 1)}{6}$.

(d) $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$.

Here's a proof by induction that does not involve a summation formula.

5.1.6 Claim. Let $n \in \mathbb{Z}_{\geq 0}$. Then $n^3 - n \equiv_3 0$.

We check the first three cases:

$$0^3 - 0 = 0 \equiv_3 0, \quad 1^3 - 1 = 0 \equiv_3 0, \quad 2^3 - 2 = 6 \equiv_3 0,$$

so the claim seems plausible. For the inductive step, we have:

Assumption: $k^3 - k \equiv_3 0$.

Goal: $(k + 1)^3 - (k + 1) \equiv_3 0$.

To deduce the goal, we start with the left side of the congruence and try to simplify it using the congruence in the assumption. We can relate the left side of the goal to the left side of the assumption by expanding the third power:

$$(k + 1)^3 - (k + 1) = k^3 + 3k^2 + 3k + 1 - k - 1 = (k^3 - k) + 3k^2 + 3k.$$

Working modulo 3, we can replace $k^3 - k$ by 0 and observe that the remaining $3k^2 + 3k$ is a multiple of 3, hence congruent to 0. Thus we are ready to write the proof.

Proof. Induction on n . The base case $n = 0$ is true since $0^3 - 0 = 0 \equiv_3 0$. Now suppose that the claim is true for some $n = k \in \mathbb{Z}_{\geq 1}$, namely that $k^3 - k \equiv_3 0$. Then

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - (k+1) = (k^3 - k) + 3k(k+1) \equiv_3 0 + 0 = 0,$$

which proves the claim when $n = k + 1$. $\square$

5.1.7 Activity. Let $n \in \mathbb{Z}_{\geq 0}$. Prove each claim by induction or disprove by giving a counterexample.

- (a) $n^2 - n \equiv_3 0$.
- (b) $6 \mid (7^n - 1)$.
- (c) $n^3 - n \equiv_6 0$.
- (d) $4^{n+2} + 5^{2n+1} \equiv_{21} 0$.

5.2 Strong induction and the fundamental theorem of arithmetic

Our goal in this section is to prove the fundamental theorem of arithmetic (4.2.6):

Theorem (Fundamental theorem of arithmetic). *Let $n \in \mathbb{Z}_{\geq 2}$. Then*

- (a) *n has a prime factorization $n = p_1 p_2 \cdots p_r$;*
- (b) *The list of primes $p_1, p_2, \dots, p_r$ is unique up to reordering.*

For the proof of (a), we will use a more powerful version of induction, called **strong induction**. Let $S(n)$ denote the statement “ n has a prime factorization”. Part (a) of the theorem states that $S(n)$ is true for all $n \in \mathbb{Z}_{\geq 2}$. The idea of the proof is as follows. If n is composite, then it has a divisor d not equal to 1 or n and thus a factorization $n = de$. Now, if we know that d and e have prime factorizations, we could combine these factorizations to get a prime factorization of n . Thus $S(d)$ and $S(e)$ together imply $S(n)$. Since d and e are less than n , we can use an enhanced version of induction in which we assume not just that the previous case holds, but that *all previous cases* hold.

5.2.1 Note (Proof by strong induction). Prove a claim $S(n)$ is true for all $n \in \mathbb{Z}_{\geq b}$ by:

- (1) “Base case”: Prove $S(b)$. (*Knock over the first domino.*)
- (2) “Inductive step”: For arbitrary $k \in \mathbb{Z}_{\geq b}$, prove that $S(n)$ for all $n \leq k$ implies $S(k+1)$. (*Show that if all the dominoes up to a certain point have fallen over, then the next one gets knocked over.*)

5.2.2 Note.

- Strong induction is more powerful than ordinary induction since the assumption in the inductive step is that the claim is true in *all previous cases* rather than just the immediately preceding case.
- Sometimes the inductive step requires $k \geq b + c$ for some constant c . In this case, it is necessary to prove multiple base cases, namely $S(b), S(b + 1), \dots, S(b + c)$.

Here is how we can use strong induction to prove (a):

Proof of (a). Strong induction on n . For the base case $n = 2$, note that 2 is prime, so we have the prime factorization $2 = 2$. Now assume that for some $k \in \mathbb{Z}_{\geq 2}$, all integers n in the range $2 \leq n \leq k$ can be written as a product of primes. Then either $k + 1$ is prime, in which case $k + 1 = (k + 1)$ is a prime factorization, or $k + 1$ is composite, in which case it has a divisor d such that $2 \leq d \leq k$. Since $d \mid (k + 1)$, there is an $e \in \mathbb{Z}$ such that $k + 1 = de$, and the bounds on d imply that $2 \leq e \leq k$ as well. By assumption, d and e can both be expressed as the product of primes; write these prime factorizations as $d = p_1 \cdots p_s$ and $e = p_{s+1} \cdots p_r$. Combining these factorizations yields $k + 1 = de = p_1 \cdots p_s p_{s+1} \cdots p_r$, showing that $k + 1$ has a prime factorization. $\square$

The proof of (b) does not require induction.

Proof of (b). To see that the prime factorization of n is unique up to reordering of the primes, suppose $n = p_1 p_2 \cdots p_r$ and $n = q_1 q_2 \cdots q_s$ are two prime factorizations of n . Canceling prime factors that appear in both factorizations, assume for contradiction that not all primes are canceled. Then the leftover primes give an equation

$$p_{i_1} p_{i_2} \cdots p_{i_u} = q_{j_1} q_{j_2} \cdots q_{j_v},$$

where no prime appears on both sides. Since p_{i_1} divides the left side, it also divides the right side. Thus, by Problem 8(c), $p_{i_1} \mid q_{j_l}$ for some l , hence $p_{i_1} = q_{j_l}$ since q_{j_l} is prime. This contradicts the fact that no prime appears on both sides of the above equation. $\square$

5.3 Problems

Problem 8. Prove each claim by induction.

(a) Let $n \in \mathbb{Z}_{\geq 1}$. Then

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1.$$

(If $n \in \mathbb{Z}_{\geq 1}$, then n **factorial**, written $n!$, is the product $1 \cdot 2 \cdots n$. For instance, $1! = 1$, $2! = 2$, and $3! = 6$.)

(b) Let $n \in \mathbb{Z}_{\geq 1}$ and let x be a real number not equal to 1. Then

$$1 + x + x^2 + \cdots + x^n = \frac{1 - x^{n+1}}{1 - x}.$$

(This is a formula for partial sums of a geometric series.)

- (c) Let $n \in \mathbb{Z}_{\geq 1}$, let $a_1, \dots, a_n \in \mathbb{Z}$, and let p be a prime. If $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i .

(Hint: use 4.2.9.)

Problem 9. Answer the question below about the following claim and proposed proof.

Claim. In any non-empty set H of horses, all the horses are the same color.

Proposed proof. Induction on n , the number of horses in H . The base case $n = 1$ holds since there is only one horse in the set. Now assume the claim is true for some $n = k \in \mathbb{Z}_{\geq 1}$, namely that any set of k horses has all horses the same color. Given a set H of $k + 1$ horses, pick two different subsets A, B of H with k horses each. By assumption, all the horses in A are the same color and all the horses in B are the same color. The intersection $A \cap B$ contains $k - 1$ horses, hence the color of the horses in A is the same as the color of the horses in B . Since $H = A \cup B$, all horses in H are the same color. This proves the claim in the case $n = k + 1$. $\square$

Are you convinced that the claim is true? If yes, explain why the claim is plausible. If not, explain why the proposed proof is not convincing.

Problem 10. Prove 4.3.1 about division with remainder using strong induction on a . You may assume $a \geq 0$. (The case $a < 0$ could be proved by a similar argument.)

(Hint: you'll need to take all elements of $\{0, 1, \dots, m - 1\}$ as the base cases.)

Problem 11. Consider the following two-player game. To start, there are two piles with the same number of objects. The players alternate turns. On each turn, a player removes any number of objects (at least one) from one of the piles. The player who removes the last object wins.

Here's an example of a game in which the initial piles each contain six objects.

○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○

Player 1 removes four objects from the second pile.

○ ○ ○ ○ ○ ○ ○ ○

Player 2 removes five objects from the first pile.

○ ○ ○

Player 1 removes one object from the first pile. That pile is now empty.

○ ○

Player 2 removes two objects from the second pile, winning the game.

A **winning strategy** in a two-player game is a complete strategy for one of the players that will ensure that player wins the game regardless of what the other player does.

- (a) For the game described above, determine whether Player 1 or Player 2 has a winning strategy and what that strategy is. Make sure the strategy works in the case where the starting piles each have n objects, for any $n \in \mathbb{Z}_{\geq 1}$.

(Hint: when analyzing strategy games of this type, start by fully analyzing the simplest cases of the game, namely the smallest values of n .)

- (b) State your result about which player has a winning strategy as a claim and prove the claim using strong induction.

Chapter 6

Functions

This section studies what may be the most important object in all of mathematics.

6.1 Definition of function

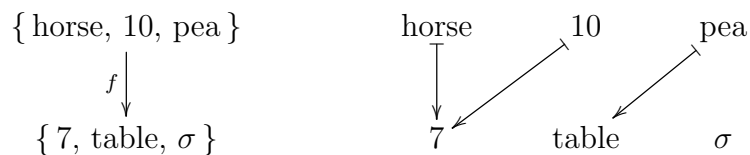
6.1.1 Definition. Let A and B be sets.

- A **function** (or **map**) f from A to B , written $f: A \rightarrow B$ or $A \xrightarrow{f} B$, is an assignment of exactly one element of B to each element of A .
- The set A is the **domain** of f and B is the **codomain** of f .
- For every $a \in A$, we denote the element of B assigned to a by $f(a)$ and say $f(a)$ is the **image** of a and a is a **preimage** of $f(a)$. We also say a **maps** to $f(a)$ and denote this by $a \mapsto f(a)$.

The standard way to define a function is to specify the sets A and B and then provide a *rule* specifying the assignment, which usually consists of a formula explaining how to calculate $f(a)$ in terms of a for all $a \in A$. We often say that a function is *given by* the rule.

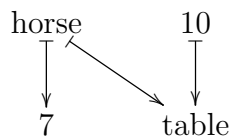
6.1.2 Example.

- The function $f: \{\text{horse}, 10, \text{pea}\} \rightarrow \{7, \text{table}, \sigma\}$ given by $\text{horse} \mapsto 7$, $10 \mapsto 7$, $\text{pea} \mapsto \text{table}$ can be visualized as



The image of horse is 7, the image of 10 is 7, and the image of pea is table. Written more succinctly, $f(\text{horse}) = 7$, $f(10) = 7$, $f(\text{pea}) = \text{table}$. The element 7 has two preimages (horse and 10), table has only one preimage (pea), and σ has no preimages.

- The assignment



is *not a function* because it assigns more than one element (7 and table) to the element horse.

- For every set A , there is a unique function $\emptyset \rightarrow A$.
- For all sets A , the **identity** function $\text{id}_A: A \rightarrow A$ is given by the rule $a \mapsto a$ for all $a \in A$.
- Let B be a set containing exactly one element. Then for all sets A , there is a unique function $A \rightarrow B$, which must map every element of A to the only element of B .
- Given $A \subseteq B$, the **inclusion** function $\iota: A \rightarrow B$ is given by $\iota(a) = a$ for all $a \in A$. The identity function id_A is the special case when $A = B$.
- $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by the rule $f(n) = n^2$
- The function $\mathbb{R} \rightarrow \mathbb{R}$ given by the rule $x \mapsto x^2$, or the function $\mathbb{R} \rightarrow [0, \infty)$ with the same rule

6.1.3 Note (Proving well-defined). For the data of sets A and B and a formula $f(a)$ to give a well-defined function, one must check the following:

- For each input $a \in A$, the rule $f(a)$ makes sense and gives exactly one output.
- For each $a \in A$, the output $f(a)$ is an element of B .

If a formula $f(a)$ fails to give a well-defined function $f: A \rightarrow B$, a common remedy is to modify the domain A by removing all elements a such that $f(a)$ does not make sense or is not in B .

6.1.4 Example.

- **Claim.** The rule $f(n) = n/2$ gives a well-defined function $f: \mathbb{Z} \rightarrow \mathbb{Z}$.

This claim is false! $1 \in \mathbb{Z}$, but $f(1) = 1/2 \notin \mathbb{Z}$.

- **Claim.** The rule $f(n) = n/2$ gives a well-defined function $f: 2\mathbb{Z} \rightarrow \mathbb{Z}$.

Proof. Let $n \in 2\mathbb{Z}$. Then $n = 2k$ for some $k \in \mathbb{Z}$, so $f(n) = f(2k) = 2k/2 = k \in \mathbb{Z}$. □

- **Claim.** The rule $f(x) = 1/x$ gives a well-defined function $f: \mathbb{R} \rightarrow \mathbb{R}$.

This claim is false! $0 \in \mathbb{R}$, but $f(0) = 1/0$ does not make sense.

- **Claim.** The rule $f(x) = 1/x$ gives a well-defined function $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$.

Proof. Let $x \in \mathbb{R}$ such that $x \neq 0$. Then $f(x) = 1/x \in \mathbb{R}$. □

6.1.5 Activity. Prove or disprove each claim.

- (a) The rule $f(n) = 3n/5$ gives a well-defined function $f: 5\mathbb{Z} \rightarrow 3\mathbb{Z}$.
- (b) The rule $f(n) = n + 2$ gives a well-defined function $5\mathbb{Z} \rightarrow 7\mathbb{Z}$.
- (c) The rule $f(x) = x + 2$ gives a well-defined function $f: [-2, \infty) \rightarrow [0, \infty)$.
- (d) The rule $f(x) = 1/(1 + x^2)$ gives a well-defined function $\mathbb{R} \rightarrow (0, 1]$.

Here is a set you have likely used many times, but may not have seen rigorously defined:

6.1.6 Definition. The **graph** of a function $f: A \rightarrow B$ is the subset of $A \times B$ defined by

$$\Gamma(f) = \{(a, b) \in A \times B : b = f(a)\}.$$

When f is a familiar function from calculus such as a polynomial, exponential, or trigonometric function, the domain and codomain are subsets of $\mathbb{R}$. Interpreting $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ as the xy -plane, f can be visualized by drawing a horizontal “real axis” for the x -values, a vertical “real axis” for the y -values, and then plotting the solution set of the equation $y = f(x)$. This solution set is *exactly the same thing* as $\Gamma(f)$.

6.2 Surjectivity

6.2.1 Definition. Let $f: A \rightarrow B$ be a function.

- The **image** (or **range**) of f is the subset of B defined as

$$\text{Im}(f) = \{b \in B : b = f(a) \text{ for some } a \in A\}.$$

- The **inverse image** of an element $b \in B$ is the subset of A defined as

$$f^{-1}(b) = \{a \in A : f(a) = b\},$$

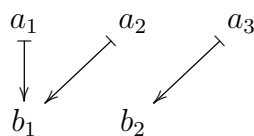
namely the set of all preimages of b .

- f is **surjective** (or **onto**) if $\text{Im}(f) = B$.

The meaning of surjectivity is that for each $b \in B$, there is at least one $a \in A$ such that $f(a) = b$. In other words, for each $b \in B$, the inverse image $f^{-1}(b)$ is non-empty.

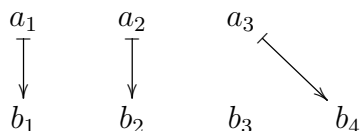
6.2.2 Example.

- The function



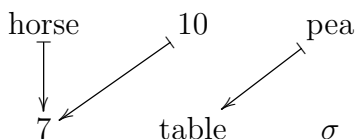
is surjective because both b_1 and b_2 have at least one preimage. More precisely, b_1 has two preimages and b_2 has one preimage.

- The function



is not surjective because b_3 has no preimage.

- Let f be the function given by



Then $\text{Im}(f) = \{7, \text{table}\}$. The inverse images of the elements of the codomain are $f^{-1}(7) = \{\text{horse}, 10\}$, $f^{-1}(\text{table}) = \{10, \text{pea}\}$, and $f^{-1}(\sigma) = \emptyset$. Since the inverse image of σ is empty, f is not surjective.

- Let g be the function that is the same as f in the previous example except that σ is removed from the codomain. Then g is surjective.
- For the function $f: \mathbb{R} \rightarrow \mathbb{R}$ with rule $f(x) = x^2$, $\text{Im}(f) = [0, \infty)$, $f^{-1}(4) = \{2, -2\}$, $f^{-1}(-4) = \emptyset$, etc. Since the inverse image of -4 is empty, f is not surjective.

Note that any non-surjective function f can be modified to be surjective by replacing the codomain of f by $\text{Im}(f)$, as in one of the examples above.

6.2.3 Note (Proving surjectivity). Let $f: A \rightarrow B$ be a function.

- To prove f is not surjective, find an element b of the codomain B and show that it has no preimage by arguing that for every $a \in A$, $f(a) \neq b$.
- To prove f is surjective, show that an arbitrary element $b \in B$ has a preimage by finding $a \in A$ such that $f(a) = b$.

Recall that $f: A \rightarrow B$ is surjective if $\text{Im}(f) = B$. While proving equality of sets usually requires two parts, here the “ $\subseteq$ ” part is clear from the definition of function, so only the “ $\supseteq$ ” part needs to be proven. However, to prove that $\text{Im}(f)$ equals any set other than the entire codomain, both parts are required.

6.2.4 Example.

- **Claim.** Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be given by $f(n) = 3n - 9$. Then $\text{Im}(f) = 3\mathbb{Z}$.

Proof. $\subseteq$: Let $m \in \text{Im}(f)$. Then $m = 3n - 9$ for some $n \in \mathbb{Z}$. Thus $m = 3(n - 3) \in 3\mathbb{Z}$.

$\supseteq$: Let $m \in 3\mathbb{Z}$. Then $m = 3k$ for some $k \in \mathbb{Z}$. Thus $m = 3(k + 3) - 9 = f(k + 3)$, so $m \in \text{Im}(f)$. $\square$

- **Claim.** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^3$. Then f is surjective.

Proof. Let $y \in \mathbb{R}$. Then the cube root of y , $\sqrt[3]{y}$, is a real number. Since $f(\sqrt[3]{y}) = (\sqrt[3]{y})^3 = y$, $y \in \text{Im}(f)$. $\square$

6.2.5 Activity.

Prove or disprove the following claims.

- (a) Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = 1/x$. Then $\text{Im}(f) = (0, \infty)$.
- (b) Let $f: 2\mathbb{Z} \rightarrow \mathbb{Z}$ be given by $f(n) = 2n$. Then $\text{Im}(f) = 4\mathbb{Z}$.
- (c) The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) = 7n - 4$ is surjective.
- (d) The function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 7x - 4$ is surjective.

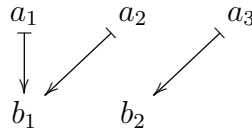
6.3 Injectivity

6.3.1 Definition. A function $f: A \rightarrow B$ is **injective** (or **one-to-one**) if f maps no two distinct elements of A to the same element of B . This means that if $a_1, a_2 \in A$ and $a_1 \neq a_2$, then $f(a_1) \neq f(a_2)$. Another common way of expressing this is by the contrapositive: if $f(a_1) = f(a_2)$, then $a_1 = a_2$. Yet another way to define injectivity of f is that every element of B has *at most one* preimage.

A function $f: A \rightarrow B$ being injective does *not* mean that the function assigns one and only one element of B to each element of A . Every function does that!

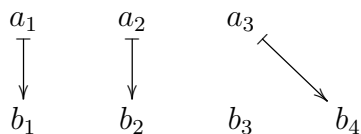
6.3.2 Example.

- The function



is not injective because b_1 has two preimages, namely a_1 and a_2 .

- The function



is injective because every b_i has at most one preimage.

6.3.3 Note (Proving injectivity). Let $f: A \rightarrow B$ be a function.

- To prove f is not injective, find two distinct elements of A that have the same image. In other words, find $a_1, a_2 \in A$ with $a_1 \neq a_2$ such that $f(a_1) = f(a_2)$.
- To prove f is injective, show that for arbitrary $a_1, a_2 \in A$ with $a_1 \neq a_2$, you must have $f(a_1) \neq f(a_2)$. Or, equivalently, prove that for any $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$, then $a_1 = a_2$.

6.3.4 Example.

- **Claim.** The function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ is injective.

This claim is false! $1 \neq -1$, but $f(1) = 1 = f(-1)$.

- **Claim.** The function $g: \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = 6x + 2$ is injective.

Proof. Suppose $g(x_1) = g(x_2)$ for some $x_1, x_2 \in \mathbb{R}$. Then $6x_1 + 2 = 6x_2 + 2$. Thus $6x_1 = 6x_2$, which implies $x_1 = x_2$. $\square$

6.3.5 Activity. Prove or disprove the following claims.

- The function $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ given by $f(x) = 3/(x-1)$ is injective.
- The function $g: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}$ given by $g(n) = n^2$ is injective.
- The function $h: \mathbb{R} \rightarrow \mathbb{R}$ given by $h(x) = x/(x^2 + 1)$ is injective.

6.4 Composition

6.4.1 Definition. Two functions $f: A \rightarrow B$ and $g: C \rightarrow D$ are **equal** if $A = C$, $B = D$, and $f(a) = g(a)$ for all $a \in A = C$.

6.4.2 Example.

- The functions $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ and $g: \mathbb{R} \rightarrow [0, \infty)$ given by $g(x) = x^2$ are *not* equal because the codomains are different.
- The functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $f(x) = (x^2 + x)/x$ and $g: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $g(x) = x + 1$ are equal since $(x^2 + x)/x = x + 1$ for all $x \neq 0$.

6.4.3 Note (Proving equality). Let $f: A \rightarrow B$ and $g: C \rightarrow D$ be functions.

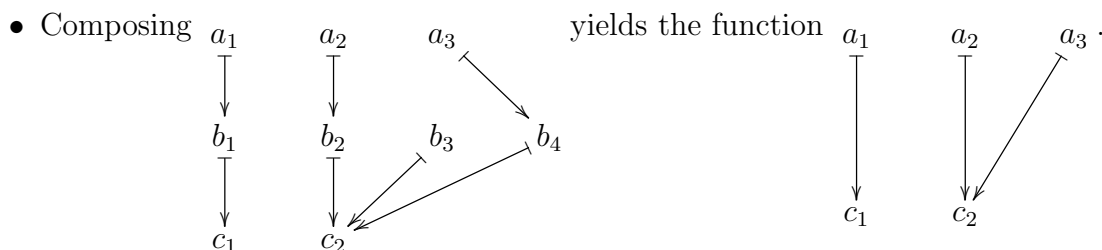
- To prove that $f = g$, check that $A = C$ and that $B = D$, and then show that $f(a) = g(a)$ for all $a \in A = C$.
- To prove that $f \neq g$, show that $A \neq C$ or that $B \neq D$, or, in the case $A = C$ and $B = D$, that there exists $a \in A = C$ such that $f(a) \neq g(a)$.

These strategies for proving or disproving two functions are equal will be used below.

One of the most useful properties of functions is the possibility of composing them.

6.4.4 Definition. Given functions $f: A \rightarrow B$ and $g: B \rightarrow C$, the **composition** of f and g is the function $g \circ f: A \rightarrow C$ given by $(g \circ f)(a) = g(f(a))$ for all $a \in A$.

6.4.5 Example.



- For any set A , $\text{id}_A \circ \text{id}_A = \text{id}_A$.
- Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be given by $f(n) = n + 3$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$ be given by $g(n) = 2n$. Then $g \circ f: \mathbb{Z} \rightarrow \mathbb{Z}$ is given by

$$(g \circ f)(n) = g(f(n)) = g(n + 3) = 2(n + 3) = 2n + 6.$$

Composing in the other order, $f \circ g: \mathbb{Z} \rightarrow \mathbb{Z}$ is given by

$$(f \circ g)(n) = f(g(n)) = f(2n) = 2n + 3.$$

Note that $f \circ g \neq g \circ f$ since $(f \circ g)(0) = 3$ while $(g \circ f)(0) = 6$.

We now explore claims about surjectivity and injectivity in the context of composition of arbitrary functions. To get a sense of whether a claim of this type is likely to be true or false, start by trying to construct a counterexample involving sets that have just a few elements each. If you are unable to find a counterexample, then the claim might be true, so try to write a proof.

6.4.6 Example.

- **Claim.** Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions and $g \circ f: A \rightarrow C$ denote the composition. If f and g are injective, then $g \circ f$ is injective.

We'll give two proofs, one for each common way of expressing injectivity.

Proof #1. Suppose $a_1, a_2 \in A$ satisfy $a_1 \neq a_2$. Then $f(a_1) \neq f(a_2)$ by the injectivity of f , so $g(f(a_1)) \neq g(f(a_2))$ by the injectivity of g . But this means $(g \circ f)(a_1) \neq (g \circ f)(a_2)$, so $g \circ f$ is injective. $\square$

Proof #2. Suppose $a_1, a_2 \in A$ satisfy $(g \circ f)(a_1) = (g \circ f)(a_2)$, namely $g(f(a_1)) = g(f(a_2))$. Then $f(a_1) = f(a_2)$ by the injectivity of g , from which it follows that $a_1 = a_2$ by the injectivity of f . Thus $g \circ f$ is injective. $\square$

- **Claim.** Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. If $g \circ f$ is surjective, then f is surjective.

This claim is false! Let $A = \{a\}$, $B = \{b_1, b_2\}$, and $C = \{c\}$ and let $f: A \rightarrow B$ be given by $f(a) = b_1$ and $g: B \rightarrow C$ be given by $g(b_1) = c$ and $g(b_2) = c$. Then $(g \circ f)(a) = g(f(a)) = g(b_1) = c$, so $g \circ f$ is surjective. But f is not surjective since b_2 has no preimage.

- **Claim.** Let $f: A \rightarrow B$, $g_1: B \rightarrow C$, and $g_2: B \rightarrow C$ be functions satisfying $g_1 \circ f = g_2 \circ f$. If f is surjective, then $g_1 = g_2$.

Proof. Let $b \in B$. Since f is surjective, $b = f(a)$ for some $a \in A$. Since $g_1 \circ f = g_2 \circ f$, we have $g_1(f(a)) = g_2(f(a))$, so $g_1(b) = g_2(b)$. Thus $g_1 = g_2$. $\square$

6.4.7 Activity. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions, and let $g \circ f: A \rightarrow C$ be the composition. Prove or disprove each claim.

- If f and g are surjective, then $g \circ f$ is surjective.
- If $g \circ f$ is surjective, then g is surjective.
- If $g \circ f$ is injective, then f is injective.
- If $g \circ f$ is injective, then g is injective.

6.4.8 Activity. Let $f_1: A \rightarrow B$, $f_2: A \rightarrow B$, and $g: B \rightarrow C$ be functions satisfying $g \circ f_1 = g \circ f_2$. Prove or disprove each claim.

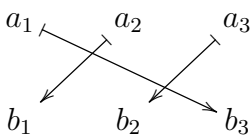
- If g is surjective, then $f_1 = f_2$.
- If g is injective, then $f_1 = f_2$.

6.5 Inverses

6.5.1 Definition. A function $f: A \rightarrow B$ is **bijective** if it is both surjective and injective. This means that every element of B has *exactly one* preimage, so f establishes a one-to-one correspondence between the elements of A and the elements of B .

6.5.2 Example.

- The function $f: A \rightarrow B$ is bijective.



- For any set A , $\text{id}_A: A \rightarrow A$ is bijective.

6.5.3 Note (Proving bijectivity). Let $f: A \rightarrow B$ be a function.

- To prove f is bijective, prove f is surjective and f is injective.
- To prove f is not bijective, prove f is not surjective or f is not injective.

6.5.4 Example.

- Claim.** The function $f: \mathbb{R} \rightarrow \mathbb{R}$ with rule $f(x) = 3x - 4$ is bijective.

Proof. Surjective: Let $y \in \mathbb{R}$. Then $(y+4)/3 \in \mathbb{R}$ and $f((y+4)/3) = 3((y+4)/3) - 4 = y + 4 - 4 = y$.

Injective: Suppose $3x_1 - 4 = 3x_2 - 4$ for $x_1, x_2 \in \mathbb{R}$. Then $3x_1 = 3x_2$, so $x_1 = x_2$. $\square$

- Claim.** The function $g: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $g(n) = 3n - 4$ is bijective.

This claim is false! $1 \in \mathbb{Z}$ but $1 \notin \text{Im}(f)$ since for all $n \in \mathbb{Z}$, $g(n) = 3n - 4 \equiv_3 -4 \equiv_3 2$, and $1 \not\equiv_3 2$. Thus g is not surjective.

6.5.5 Activity. Prove or disprove the following claims.

- The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) = n + 3$ is bijective.
- The function $g: \mathbb{Z} \rightarrow 2\mathbb{Z}$ given by $g(n) = 2n - 6$ is bijective.

6.5.6 Note (Bijectivity and graphs). Determine whether a function $f: \mathbb{R} \rightarrow \mathbb{R}$ with a “nice” rule is most likely injective, surjective, and bijective by looking at the graph $\Gamma(f)$:

- f is injective if and only if every horizontal line in $\mathbb{R}^2$ intersects $\Gamma(f)$ in *at most one* point (we say $\Gamma(f)$ satisfies the “horizontal line test”).
- f is surjective if and only if every horizontal line in $\mathbb{R}^2$ intersects $\Gamma(f)$ in *at least one* point.

- Thus f is bijective if and only if every horizontal line in $\mathbb{R}^2$ intersects $\Gamma(f)$ in *exactly one* point.

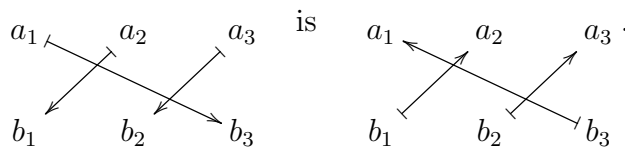
As graphs are imprecise and can be misleading if the rule of f is not “nice”, looking at a graph can provide useful intuition but is not a rigorous way to prove that a function is bijective.

The following definition is closely related to bijectivity.

6.5.7 Definition. Two functions $f: A \rightarrow B$ and $g: B \rightarrow A$ are **inverse** functions if $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. We also say f is **invertible** and that g is the **inverse** of f , and we denote g by f^{-1} .

6.5.8 Example.

- The inverse of



- The inverse of id_A is id_A .

The first example demonstrates a key idea: given any bijective function, reversing the direction of the mapping yields the inverse function. This is proved in 6.5.13 below.

6.5.9 Note (Proving invertibility).

- For a function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by a rule $f(x)$, the way to find an inverse of f , if it exists, is to set $y = f(x)$ and solve for x in terms of y , obtaining $x = g(y)$. If this is possible and if $g(y)$ gives a well-defined function $g: \mathbb{R} \rightarrow \mathbb{R}$, then check that the compositions $g \circ f$ and $f \circ g$ are equal to the identity function $\text{id}_{\mathbb{R}}$. A similar procedure works for functions on other sets where we can “solve for” the input variable, such as $\mathbb{Z}$.
- To prove that a function $f: A \rightarrow B$ is invertible, just define an appropriate function $g: B \rightarrow A$ and show that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. There is no need to explain how to derive g .
- To prove that a function $f: A \rightarrow B$ is not invertible, show that f is not injective or f is not surjective. This works because f is invertible if and only if f is bijective (see 6.5.13).

6.5.10 Example.

- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x - 4$. Let's try to find an inverse for f . Set $y = 3x - 4$ and solve for x , which yields $x = (y + 4)/3$. This formula makes sense for all y , so $g: \mathbb{R} \rightarrow \mathbb{R}$ given by $g(y) = (y + 4)/3$ should be the inverse function.

Claim. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 3x - 4$ is invertible.

Proof. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(y) = (y + 4)/3$. Then

$$(g \circ f)(x) = g(f(x)) = g(3x - 4) = (3x - 4 + 4)/3 = x = \text{id}_{\mathbb{R}}(x) \quad \text{and} \\ (f \circ g)(y) = f(g(y)) = f((y + 4)/3) = 3(y + 4)/3 - 4 = y = \text{id}_{\mathbb{R}}(y),$$

so $g \circ f = \text{id}_{\mathbb{R}}$ and $f \circ g = \text{id}_{\mathbb{R}}$. Thus g is the inverse of f . $\square$

- Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be given by $f(n) = 2n$. Let's try to find an inverse. Setting $m = 2n$ and solving for n , we get $n = m/2$. But $g(m) = m/2$ does not give a well-defined function on $\mathbb{Z}$.

Claim. The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) = 2n$ is invertible.

This claim is false! Since $f(n) = 2n \in 2\mathbb{Z}$ for all n , $\text{Im}(f) \subseteq 2\mathbb{Z}$. Since 1 is odd, it is not in $\text{Im}(f)$, so f is not surjective.

6.5.11 Activity. Determine the inverse of the function. State your result as a claim and prove the claim.

(a) $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R} \setminus \{0\}$ given by $f(x) = 1/(1 - x)$.

(b) $g: \mathbb{Z} \rightarrow 3\mathbb{Z}$ given by $g(n) = 3n$.

6.5.12 Activity.

- (a) Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ are invertible functions. Prove that $g \circ f$ is invertible by determining its inverse in terms of f^{-1} and g^{-1} and proving the compositions of $g \circ f$ and this inverse equal id_A and id_C .
- (b) Prove or disprove: if $f: A \rightarrow B$ and $g: B \rightarrow A$ satisfy $g \circ f = \text{id}_A$, then f and g are inverse functions.

As mentioned above, invertible functions are exactly the same as bijective functions.

6.5.13 Proposition. A function $f: A \rightarrow B$ is invertible if and only if it is bijective.

Proof. $\implies$: Let $g: B \rightarrow A$ be the inverse of f . Suppose $f(a_1) = f(a_2)$ for $a_1, a_2 \in A$. Then

$$a_1 = \text{id}_A(a_1) = (g \circ f)(a_1) = g(f(a_1)) = g(f(a_2)) = (g \circ f)(a_2) = \text{id}_A(a_2) = a_2,$$

so f is injective. To see that f is surjective, suppose $b \in B$. Then $g(b) \in A$ and

$$f(g(b)) = (f \circ g)(b) = \text{id}_B(b) = b,$$

so f is surjective. Thus f is bijective.

$\impliedby$: Define $B \rightarrow A$ to be the function that assigns to each element $b \in B$ the unique element $a \in A$ such that $f(a) = b$. Such an element a exists since f is surjective, and is unique since f is injective. Now we use this definition of g to compute

$$(g \circ f)(a) = g(f(a)) = a = \text{id}_A(a), \quad (f \circ g)(b) = f(g(b)) = b = \text{id}_B(b).$$

Thus $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$, so g is the inverse of f . $\square$

6.6 Problems

Problem 12. Let A and B be sets. Suppose there exists an injective function $f: A \rightarrow B$ and an injective function $g: B \rightarrow A$. Prove that there exists a bijective function $h: A \rightarrow B$.

(Warning: this problem is very challenging.)

Problem 13. Suppose A is a set. Prove that there is no surjective function from A to $\mathcal{P}(A)$.

(Hint: given any function from A to the power set $\mathcal{P}(A)$, use an idea similar to the diagonalization technique in the proof of 3.5.8 to define a subset $S \subseteq A$ that is not in the image of the function.)

Chapter 7

Relations

Relations are a generalization of functions, expressing general relationships between the elements of two sets.

7.1 Definition of relation

A (binary) relation expresses a property satisfied by some elements of a pair of sets A and B .

7.1.1 Definition. A **relation** $\sim$ from A to B specifies, for all $a \in A$ and $b \in B$, whether a is **related** to b , denoted $a \sim b$. If a is not related to b , we write $a \not\sim b$. If $A = B$, then we say $\sim$ is a relation on A .

7.1.2 Example. Every function $f: A \rightarrow B$ can be viewed as a relation from A to B in which each $a \in A$ is related to $f(a) \in B$.

7.1.3 Example. In previous chapters, we worked with many relations that have standard names and a standard symbol that is used in place of $\sim$.

- “*Equals*”: $a = b$ for $a, b \in A$ (here A denotes a fixed set)
- “*Less than*”: $x < y$ for $x, y \in \mathbb{R}$
- “*Greater than or equal to*”: $x \geq y$ for $x, y \in \mathbb{R}$
- “*Divides*”: $a \mid b$ for $a, b \in \mathbb{Z}$
- “*Congruent modulo m* ”: $a \equiv_m b$ for $a, b \in \mathbb{Z}$ (here m denotes a fixed positive integer)
- “*Subset*”: $A \subseteq B$ for A, B in the “set” $\mathcal{S}$ of all sets
- “*Implies*”: $p \implies q$ for p, q in the “set” $\mathcal{P}$ of all mathematical statements
- “*Element of*”: $x \in A$ for x in a “universal set” and A in the “set” $\mathcal{S}$ of all sets

The notion of graph of a function generalizes to relations.

7.1.4 Definition. Let $\sim$ be a relation from A to B . The **graph** of $\sim$, denoted $\Gamma(\sim)$, is the subset of $A \times B$ defined as

$$\Gamma(\sim) = \{(a, b) \in A \times B : a \sim b\}.$$

7.1.5 Activity. Sketch the graph of the following relations.

- “Equals” on $\mathbb{R}$
- “Less than” on $\mathbb{R}$
- “Greater than or equal to” on $\mathbb{R}$
- “Divides” on $\mathbb{Z}$
- “Congruent modulo 2” on $\mathbb{Z}$

Viewing Γ as a way to associate a set to each relation (see also Problem 14), one can define set-theoretic notions like union, intersection, complement, and difference for relations. Can you guess which relation on $\mathbb{R}$ is the complement of “less than”? Which relation on $\mathbb{R}$ is the union of “equals” and “less than”?

7.2 Equivalence relations

7.2.1 Definition. Let $\sim$ denote a relation on A .

- $\sim$ is **reflexive** if for all $a \in A$, $a \sim a$;
- $\sim$ is **symmetric** if for all $a, b \in A$, if $a \sim b$, then $b \sim a$;
- $\sim$ is **transitive** if for all $a, b, c \in A$, if $a \sim b$ and $b \sim c$, then $a \sim c$.

A relation that is reflexive, symmetric, and transitive is an **equivalence relation**.

Note that to prove $\sim$ is not reflexive, not symmetric, or not transitive, it suffices to provide a single counterexample.

7.2.2 Example.

- **Claim.** The relation “less than or equal to” on $\mathbb{R}$ is reflexive and transitive but not symmetric.

Proof. Reflexive: for all $x \in \mathbb{R}$, $x \leq x$.

Not symmetric: $1 \leq 2$ but $2 \not\leq 1$.

Transitive: if $x \leq y$ and $y \leq z$, then $x \leq z$. □

- **Claim.** Let $m \in \mathbb{Z}_{\geq 1}$. The relation “congruent modulo m ” on $\mathbb{Z}$ is an equivalence relation.

Proof. Reflexive: for all $a \in \mathbb{Z}$, $a \equiv_m a$ since $a = a + 0m$.

Symmetric: if $a \equiv_m b$, then $a = b + mk$ for some $k \in \mathbb{Z}$. Thus $b = a - mk$, so $b \equiv_m a$. Thus the relation is symmetric.

Transitive: if $a \equiv_m b$ and $b \equiv_m c$, then $a = b + mj$ and $b = c + mk$ for some $j, k \in \mathbb{Z}$. Thus

$$a = b + mj = (c + mk) + mj = c + m(k + j),$$

so $a \equiv_m c$. □

7.2.3 Activity. For each relation, determine whether it is reflexive, whether it is symmetric, and whether it is transitive. Is the relation an equivalence relation?

- “Equals” on a set A
- “Less than” on $\mathbb{R}$
- “Greater than or equal to” on $\mathbb{R}$
- “Divides” on $\mathbb{Z}$
- The relation $\sim$ on $\mathbb{R}$ given by $x \sim y$ if $x^2 = y^2$
- The relation $\sim$ on $\mathbb{Z}$ given by $a \sim b$ if a and b are both even
- The relation $\sim$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$ given by $(x_1, y_1) \sim (x_2, y_2)$ if $x_1 y_2 = x_2 y_1$.

7.2.4 Definition. Let $\sim$ be an equivalence relation on A . The **equivalence class** of $a \in A$ is the subset of A defined by

$$[a] = \{ b \in A : a \sim b \}.$$

7.2.5 Example. For congruence modulo 2 on $\mathbb{Z}$, $[0]$ is the set of all even integers and $[1]$ is the set of all odd integers. Note that $[0] = [2]$ and, more generally, any two even numbers define the same equivalence class. Similarly, any two odd numbers define the same equivalence class.

7.2.6 Proposition. Suppose $\sim$ is an equivalence relation on A .

- For all $a \in A$, $a \in [a]$.
- For all $a, b \in A$, either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Proof. (a): This is immediate by reflexivity.

(b): Suppose $[a] \cap [b]$ is nonempty, hence the intersection contains some $c \in A$. Then $a \sim c$ and $b \sim c$. By symmetry, $c \sim a$, and by transitivity, $b \sim a$.

We claim that $[a] = [b]$. Let $d \in [a]$. Then $a \sim d$. Since $b \sim a$, by transitivity, $b \sim d$. Thus $d \in [b]$, so $[a] \subseteq [b]$. The proof that $[b] \subseteq [a]$ is similar. □

The proposition implies that every $a \in A$ is contained in exactly one equivalence class. Another way to say this is that A is the disjoint union of equivalence classes.

7.2.7 Definition. Let $\sim$ be an equivalence relation on A . The set of all equivalence classes, denoted

$$A/\sim = \{ [a] : a \in A \},$$

is the **quotient set** of A by $\sim$. The function $\pi: A \rightarrow A/\sim$ given by $\pi(a) = [a]$ is the **quotient map**.

7.2.8 Note.

- The *elements* of $A/\sim$ are *sets*.
- The quotient set can be tricky to work with because the description of one of its elements as $[a]$ is non-unique in the sense that $[a] = [b]$ if and only if $a \sim b$. In other words, an element of $A/\sim$ can be expressed as the equivalence class of any one of the elements it contains.
- The quotient map π is always surjective since for any $[a] \in A/\sim$, $\pi(a) = [a]$. Moreover, the preimage $\pi^{-1}([a])$ is equal to $[a]$.

7.2.9 Example. Consider congruence modulo m on $\mathbb{Z}$. The quotient set

$$\mathbb{Z}/\equiv_m = \{ [0], [1], \dots, [m-1] \}$$

is in bijection with the set $\{0, 1, \dots, m-1\}$ of possible remainders modulo m . The quotient map $\pi: \mathbb{Z} \rightarrow \mathbb{Z}/\equiv_m$ maps each integer to the equivalence class of its remainder modulo m .

Note that the natural-looking rule $[n] \mapsto n$ does not give a well-defined function $\mathbb{Z}/\equiv_m \rightarrow \mathbb{Z}$ because each element of the domain is mapped to infinitely many elements of the codomain! For instance, the rule states that $[0] \mapsto 0$ and $[m] \mapsto m$, but $[0]$ and $[m]$ are the same element of the domain.

7.2.10 Activity. Let $f: A \rightarrow B$ be a function. Define a relation $\sim_f$ on A by

$$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2).$$

- Prove that $\sim_f$ is an equivalence relation.
- Prove that f is injective if and only if the quotient map $\pi: A \rightarrow A/\sim_f$ is bijective.
- For all $a \in A$, every element of $[a]$ has the same image under f . Explain why this shows that the rule $\bar{f}([a]) = f(a)$ gives a well-defined function $\bar{f}: A/\sim_f \rightarrow B$.
(Hint: you need to make sure that the definition of $\bar{f}$ does not depend on which choice of equivalence class is used to represent an element of $A/\sim$.)
- Prove that $\bar{f}$ is injective.
- Prove that the composition $\bar{f} \circ \pi$ is equal to f .

7.3 Problems

Problem 14. Fixing two sets A and B , the assignment $\sim \mapsto \Gamma(\sim)$ (see 7.1.4) gives a function

$$\Gamma: \{ \sim: \sim \text{ is a relation from } A \text{ to } B \} \rightarrow \mathcal{P}(A \times B)$$

from the set of all relations from A to B to the set of all subsets of $A \times B$ (the power set of $A \times B$). Prove that Γ is bijective.

Problem 15. Consider the relation $\sim$ on $\mathbb{Z} \times \mathbb{Z}_{\geq 1}$ given by

$$(a, b) \sim (c, d) \text{ if } ad = bc.$$

(a) Prove that $\sim$ is an equivalence relation.

Now consider the function $q: \mathbb{Z} \times \mathbb{Z}_{\geq 1} \rightarrow \mathbb{R}$ given by $q(a, b) = a/b$.

(b) Prove that $(a, b) \sim (c, d)$ if and only if $q(a, b) = q(c, d)$.

(c) Part (b) shows that $\sim$ is the same as $\sim_q$, where $\sim_q$ is defined in 7.2.10. Thus q induces a bijection between $(\mathbb{Z} \times \mathbb{Z}_{\geq 1})/\sim$ and the image of q . Thus, using a nice description of the image of q , find a nice description of $(\mathbb{Z} \times \mathbb{Z}_{\geq 1})/\sim$.

Problem 16. Consider the relation $\sim$ on $\mathbb{R}^2$ given by

$$(x_1, y_1) \sim (x_2, y_2) \text{ if } y_2 - y_1 = x_2 - x_1.$$

(a) Prove that $\sim$ is an equivalence relation.

(b) Describe the equivalence classes $[(0, 0)]$ and $[(0, 1)]$ as solution sets of equations. As these are subsets of $\mathbb{R}^2$, also provide a geometric description of these subsets.

(c) Find a geometric description of the quotient set $\mathbb{R}^2/\sim$.

Problem 17. Let $\mathcal{S}$ denote the “set” containing all sets. Consider the relation $\sim$ given by

$$A \sim B \text{ if there exists a bijection } f: A \rightarrow B.$$

(a) Prove that $\sim$ is an equivalence relation.

The **cardinality** of A , denoted $|A|$, is the equivalence class $[A] \in \mathcal{S}/\sim$. (Using the notation of the quotient map $\pi: \mathcal{S} \rightarrow \mathcal{S}/\sim$, $|A| = \pi(A)$.) Thus $|A| = |B|$ if and only if there exists a bijection $f: A \rightarrow B$. The cardinalities of some common sets have a special notation:

- $|\{ \} |$ is denoted 0
- $|\mathbb{Z}|$ is denoted $\aleph_0$
- $|\{ 1, 2, \dots, n \}|$ is denoted n
- $|\mathbb{R}|$ is denoted $\mathfrak{c}$

Thus we can view $\mathbb{Z}_{\geq 0} \cup \{ \aleph_0, \mathfrak{c} \}$ as a subset of $\mathcal{S}/\sim$.

Consider the relation $\leq$ on $\mathcal{S}/\sim$ given by

$$|A| \leq |B| \text{ if there exists an injective function } f: A \rightarrow B.$$

- (b) Determine whether $\leq$ is reflexive, whether it is symmetric, and whether it is transitive. Is $\leq$ an equivalence relation?
- (c) Prove that $n \leq \aleph_0$ and that $\aleph_0 \leq \mathfrak{c}$.
- (d) Prove that $|A| \leq |\mathcal{P}(A)|$ for all sets A .

We say $|A| < |B|$ if $|A| \leq |B|$ and $|A| \neq |B|$. Thus, combining part (c) with 3.5.7 and 3.5.8, we see that

$$n < \aleph_0 < \mathfrak{c}.$$

Similarly, combining (d) with Problem 13 shows that

$$|A| < |\mathcal{P}(A)| < |\mathcal{P}(\mathcal{P}(A))| < |\mathcal{P}(\mathcal{P}(\mathcal{P}(A)))| < \dots,$$

which implies (for instance by taking $A = \mathbb{Z}$) that there are infinitely many distinct cardinalities of infinite sets.

Chapter 8

Limits

Limits provide a rigorous understanding of some notions related to infinity. Given an infinite list of numbers, what can be said about the trend of the numbers arbitrarily deep in the list? What can be said about the value of a function as the inputs become arbitrarily close to a given number?

8.1 Absolute value and inequalities

The **absolute value** function $|\cdot|: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is given by the rule

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Geometrically, $|x|$ can be interpreted as the distance between x and 0 on the number line. Since distance is multiplicative, $|xy| = |x| \cdot |y|$. More generally, if $x, y \in \mathbb{R}$, then

$$|x - y| = |y - x| = \text{distance between } x \text{ and } y \text{ on the number line.}$$

To make a vague statement like “ x and y are close” rigorous, we can use an inequality like

$$|x - y| < d,$$

which means that the distance between x and y is less than d , where d is a positive real number.

8.1.1 Note. It will be useful to recall the following basic facts related to inequalities. Suppose $w, x, y, z, c \in \mathbb{R}$.

- (a) If c is positive, then $x < y \iff cx < cy$.
- (b) If c is negative, then $x < y \iff cx > cy$.
- (c) If x, y are positive, then $x < y \iff 1/x > 1/y$.

(d) If $x < y$ and $w < z$, then $x + w < y + z$.

Analogous facts are true with $<$ replaced by $\leq$.

Having reviewed basic facts about inequalities, we can now prove a useful fact stating that the above absolute value inequality is equivalent to a compound inequality.

8.1.2 Lemma. *Let $x, y \in \mathbb{R}$ and $d \in \mathbb{R}_{\geq 0}$. Then*

$$|x - y| < d \iff -d < x - y < d.$$

Proof. Suppose $x - y \geq 0$. Then $x - y > -d$ is always true. Moreover, $x - y = |x - y|$, so $|x - y| < d \iff x - y < d$, which gives the result. Now suppose instead that $x - y < 0$. Then $x - y < d$ is always true. Moreover, $|x - y| = -(x - y)$, so $|x - y| < d \iff -(x - y) < d \iff x - y > -d$, giving the result. $\square$

Note that adding y to each expression in the compound inequality $-d < x - y < d$ yields $y - d < x < y + d$, which can also be interpreted as $x \in (y - d, y + d)$.

The following useful inequality derives its name from a more general statement about lengths of vectors that can be interpreted as the sides of a triangle.

8.1.3 Proposition (Triangle inequality). *Let $x, y \in \mathbb{R}$. Then*

$$|x + y| \leq |x| + |y|.$$

Proof. It suffices to prove the compound inequality $-(|x| + |y|) \leq x + y \leq |x| + |y|$. This follows from adding the inequalities $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. $\square$

8.2 Limit of a sequence

Recall from 3.4.4 that a sequence (of real numbers) is an element of $\mathbb{R}^\infty$. Namely, a sequence is an infinite list $(a_1, a_2, a_3, \dots)$, where each $a_n \in \mathbb{R}$. Each number in the list is called a **term** of the sequence. A sequence can be denoted more concisely as (a_n) with the understanding that n takes on all values in the index set $\mathbb{Z}_{\geq 1}$.

8.2.1 Definition. Let (a_n) denote a sequence and $a \in \mathbb{R}$. We say that the **limit** of (a_n) is a , denoted $\lim a_n = a$, if for every $\epsilon > 0$, there is a real number N such that $|a_n - a| < \epsilon$ for all $n > N$.

Intuitively, “the limit of (a_n) is a ” means that a_n can be made arbitrarily close to a by taking n sufficiently large. The absolute value inequality with a small real number ϵ makes precise the idea of “arbitrarily close to”, and the ordinary inequality with a large real number N captures the idea of “sufficiently large”.

8.2.2 Note (How to analyze $\lim a_n$).

- *Step 1:* Guess the limit a by looking at the behavior of a_n as n gets large.

- *Step 2:* Use algebra and properties of inequalities to find a simple upper bound for $|a_n - a|$ that depends on n .
- *Step 3:* Set this upper bound equal to ϵ and solve for n to obtain N (this will depend on ϵ).
- *Step 4:* State and prove the claim that $\lim a_n = a$ using the value of N found in Step 3 and the algebraic techniques and inequalities in Step 2.

8.2.3 Example. Consider the sequence $(1/n)$. Since $1/n$ gets closer and closer to 0 as n increases, we can guess that $\lim 1/n = 0$. To prove this, we need to show that for every ϵ , there exists N such that $|1/n - 0| < \epsilon$ for all $n > N$. Since $|1/n - 0| = 1/n$, we need $1/n < \epsilon$. Solving $1/n = \epsilon$ yields $n = 1/\epsilon$, so we take $N = 1/\epsilon$. We now use the ideas in this scratch work to state and prove the claim.

Claim. $\lim 1/n = 0$.

Proof. Given $\epsilon > 0$, let $N = 1/\epsilon$. Then for all $n > N$,

$$|1/n - 0| = 1/n < 1/N = \epsilon. \quad \square$$

8.2.4 Example. Consider the sequence $(\sqrt{4 + 1/n})$. As $\lim 1/n = 0$, we guess that $\lim \sqrt{4 + 1/n} = \sqrt{4} = 2$. To prove this, we need to show that for every ϵ , there exists N such that $|\sqrt{4 + 1/n} - 2| < \epsilon$ for all $n > N$. Note that $|\sqrt{4 + 1/n} - 2| = \sqrt{4 + 1/n} - 2$. This is still difficult to work with, as n appears inside the square root. Rationalizing this expression by multiplying by $(\sqrt{4 + 1/n} + 2)/(\sqrt{4 + 1/n} + 2)$, we see that

$$\sqrt{4 + 1/n} - 2 = \frac{4 + 1/n - 4}{\sqrt{4 + 1/n} + 2} = \frac{1/n}{\sqrt{4 + 1/n} + 2} < \frac{1/n}{\sqrt{4} + 2} = \frac{1}{4n}.$$

Solving $1/(4n) = \epsilon$, we get $n = 1/(4\epsilon)$, so we take $N = 1/(4\epsilon)$. We are now ready to write the proof.

Claim. $\lim \sqrt{4 + 1/n} = 2$.

Proof. Given $\epsilon > 0$, let $N = 1/(4\epsilon)$. Then for all $n > N$,

$$|\sqrt{4 + 1/n} - 2| = \sqrt{4 + 1/n} - 2 = \frac{4 + 1/n - 4}{\sqrt{4 + 1/n} + 2} = \frac{1/n}{\sqrt{4 + 1/n} + 2} < \frac{1}{4n} < \frac{1}{4N} = \epsilon. \quad \square$$

8.2.5 Activity. Guess the limit of the sequence. Write your guess as a claim, and then prove the claim.

(a) $(1/n^2)$

(b) $\frac{2n - 1}{3n + 1}$

(c) $(-1)^n/n$

(d) $\frac{n}{n^3 + 4}$

(e) $\sqrt{n+1} - \sqrt{n}$

Not every sequence has a limit. To distinguish those sequences that have a limit from those that do not, we make the following definition.

8.2.6 Definition. Let (a_n) be a sequence. If there exists $a \in \mathbb{R}$ such that $\lim a_n = a$, then we say (a_n) **converges**. Otherwise, (a_n) **diverges**.

8.2.7 Example. Consider the sequence $((-1)^n)$. The terms alternate between -1 and 1 . Since the terms do not approach a single real number, we can guess that the sequence diverges. To prove that a sequence diverges, we have to show that for every $a \in \mathbb{R}$, there exists $\epsilon > 0$ such that for all N , $|a_n - a| \geq \epsilon$ for some $n > N$. Since no number is within 1 of both -1 and 1 , we can take $\epsilon = 1$.

Claim. *The sequence $((-1)^n)$ diverges.*

Proof. Given $a \in \mathbb{R}$, let $\epsilon = 1$. Then for all N , the terms a_n with $n > N$ take on the values of -1 and 1 . Note that

$$|a_n - a| = \begin{cases} |1 - a| & \text{if } n \text{ is even} \\ |-1 - a| & \text{if } n \text{ is odd} \end{cases}$$

If $a \geq 0$, then $|-1 - a| \geq 1 = \epsilon$. If $a < 0$, then $|1 - a| > 1 = \epsilon$. Thus for any a and N , there is a_n with $n > N$ such that $|a_n - a| \geq 1$. $\square$

8.3 Limit laws

So far, we have analyzed limits of particular sequences using the definition of limit. To deepen our understanding of limits, we will prove general properties that apply to limits of arbitrary sequences.

8.3.1 Lemma. $\lim a_n = a \iff \lim |a_n - a| = 0$.

Proof. This follows immediately from the definition of limit. $\square$

The next statement captures the idea of Steps 2 and 3 in 8.2.2.

8.3.2 Lemma. *If $|a_n - a| \leq b_n$ for all n and $\lim b_n = 0$, then $\lim a_n = a$.*

Proof. Given $\epsilon > 0$, since $\lim b_n = 0$, there is N such that $|b_n| < \epsilon$ for all $n > N$. Note that $|b_n| = b_n$ since $b_n \geq |a_n - a| \geq 0$. Thus, for all $n > N$,

$$|a_n - a| \leq b_n = |b_n| < \epsilon. \quad \square$$

8.3.3 Definition. A sequence (a_n) is **bounded** if there exists m such that $|a_n| \leq m$ for all n .

Note that the inequality $|a_n| \leq m$ means that $-m \leq a_n \leq m$, so a sequence is bounded if there is a closed interval that contains all terms of the sequence.

8.3.4 Lemma. *If (a_n) converges, then (a_n) is bounded.*

Proof. Since (a_n) converges, $\lim a_n = a$ for some $a \in \mathbb{R}$. Taking $\epsilon = 1$, there exists N such that $|a_n - a| < 1$ for all $n > N$. Let m be the maximum of $|a| + 1$ and all $|a_n|$ such that $n \leq N$. Then $|a_n| \leq m$ for all n , so (a_n) is bounded. $\square$

8.3.5 Lemma. *If $\lim a_n = 0$ and (b_n) is bounded, then $\lim a_n b_n = 0$.*

8.3.6 Activity. Prove the lemma.

These lemmas are useful for proving properties about limits of sequences built from other sequences using basic operations.

8.3.7 Theorem (Limit laws). *Suppose $\lim a_n = a$, $\lim b_n = b$, $c \in \mathbb{R}$, and $k \in \mathbb{Z}_{\geq 0}$. Then*

- (a) $\lim ca_n = ca$;
- (b) $\lim(a_n + b_n) = a + b$;
- (c) $\lim a_n b_n = ab$;
- (d) $\lim a_n/b_n = a/b$ if $b \neq 0$ and $b_n \neq 0$ for all n ;
- (e) $\lim a_n^k = a^k$;
- (f) $\lim a_n^{1/k} = a^{1/k}$ if $a_n \geq 0$ for all n .

We provide a proof of the first three parts of the theorem. The remaining parts can be proved using similar techniques.

Proof of (a)-(c). (a): If $c = 0$, then the claim is trivial, so suppose $c \neq 0$. Given $\epsilon > 0$, then since $\lim a_n = a$, there is N such that $|a_n - a| < \epsilon/|c|$ for all $n > N$. Then for all $n > N$,

$$|ca_n - ca| = |c(a_n - a)| = |c| \cdot |a_n - a| < \epsilon.$$

(b): Since $\lim a_n = a$, there is N_1 such that for all $n > N_1$, $|a_n - a| < \epsilon/2$. Similarly, since $\lim b_n = b$, there is N_2 such that for all $n > N_2$, $|b_n - b| < \epsilon/2$. Let $N = \max\{N_1, N_2\}$. Then for all $n > N$,

$$|a_n + b_n - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \epsilon/2 + \epsilon/2 = \epsilon.$$

(c): Note that by the triangle inequality,

$$|a_n b_n - ab| = |(a_n b_n - ab_n) + (ab_n - ab)| \leq |b_n| \cdot |a_n - a| + |a| \cdot |b_n - b|.$$

Since (b_n) converges, (b_n) is bounded by 8.3.4. By 8.3.5, each term on the right defines a sequence whose limit is 0. Thus, by (b), the right side converges to 0, so 8.3.2 implies that $\lim |a_n b_n - ab| = 0$ as well. $\square$

8.4 Limits of a function

In this section, we consider functions $f: D \rightarrow \mathbb{R}$, where the domain D is a subset of $\mathbb{R}$.

8.4.1 Definition. Let $a \in \mathbb{R}$ and $L \in \mathbb{R}$. We say that the **limit** of f as x approaches a is L , denoted $\lim_{x \rightarrow a} f(x) = L$, if for every $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \text{ for all } x \neq a \text{ such that } |x - a| < \delta.$$

If there is no such L , then we say that $\lim_{x \rightarrow a} f(x)$ does not exist.

8.4.2 Note.

- Intuitively, “the limit of f as x approaches a is L ” means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a .
- Implicit in the definition of limit is the requirement that $f(x)$ is well-defined for all $x \neq a$ such that $|x - a| < \delta$. This is only possible if the domain D contains a set of the form $(a - \delta, a) \cup (a, a + \delta)$.

8.4.3 Note (How to analyze $\lim_{x \rightarrow a} f(x)$).

- *Step 1:* Guess the limit L by looking at the behavior of f as x gets close to a .
- *Step 2:* Use algebra and inequalities to find a simple upper bound for $|f(x) - L|$ of the form $h(x) \cdot |x - a|$.
- *Step 3:* By imposing a condition $|x - a| < \delta_1$ if necessary, find a constant upper bound for $h(x)$.
- *Step 4:* Set this constant times $|x - a|$ equal to ϵ and solve for $|x - a|$ to obtain δ_2 .
- *Step 5:* State and prove the claim that $\lim_{x \rightarrow a} f(x) = L$ using $\delta = \min\{\delta_1, \delta_2\}$ and the algebraic techniques and inequalities in Steps 2 and 3.

8.4.4 Example. Consider $\lim_{x \rightarrow 3} x^2$. As x gets close to 3, x^2 approaches $3^2 = 9$. Now, note that

$$|x^2 - 9| = |(x + 3)(x - 3)| = |x + 3| \cdot |x - 3|.$$

We can find an upper bound for $|x + 3|$ by ensuring x is close to 3. For instance, if $|x - 3| < 1$, then $2 < x < 4$, so $|x + 3| < 7$. Now, setting $7|x - 3| = \epsilon$, we get $|x - 3| = \epsilon/7$, so we take $\delta_2 = \epsilon/7$. We'll use $\delta = \min\{1, \epsilon/7\}$ in the proof.

Claim. $\lim_{x \rightarrow 3} x^2 = 9$.

Proof. Given $\epsilon > 0$, let $\delta = \max\{1, \epsilon/7\}$. Then for all x such that $x \neq 3$ and $|x - 3| < \delta$, we have $2 < x < 4$, so

$$|x^2 - 9| = |(x + 3)(x - 3)| = |x + 3| \cdot |x - 3| < 7 \cdot |x - 3| < 7 \cdot \delta \leq \epsilon. \quad \square$$

8.4.5 Example. Consider $\lim_{x \rightarrow 0} x \sin(1/x)$. As x approaches 0, $1/x$ gets large, so $\sin(1/x)$ continues to oscillate, taking on values between -1 and 1 . Since sine is bounded and the factor of x approaches 0, we can guess that $x \sin(1/x)$ approaches 0 as well. Using the fact that $|\sin(1/x)| \leq 1$, note that

$$|x \sin(1/x) - 0| = |\sin(1/x)| \cdot |x| \leq |x|,$$

so we can take $\delta = \epsilon$.

Claim. $\lim_{x \rightarrow 0} x \sin(1/x) = 0$.

Proof. Given $\epsilon > 0$, let $\delta = \epsilon$. Then for all x such that $x \neq 0$ and $|x| < \delta$,

$$|x \sin(1/x) - 0| = |\sin(1/x)| \cdot |x| \leq |x| < \delta = \epsilon. \quad \square$$

8.4.6 Activity. Guess the limit. Write your guess as a claim, and then prove the claim.

- (a) $\lim_{x \rightarrow 5} (x^2 - 3x)$
- (b) $\lim_{x \rightarrow 2} x^3$
- (c) $\lim_{x \rightarrow 0} x^2 \cos(1/x)$
- (d) $\lim_{x \rightarrow a} 1/x$ for all $a \neq 0$
- (e) $\lim_{x \rightarrow a} \sqrt{x}$ for all $a > 0$

Limits of functions have many of the same properties as limits of sequences. The proof of the limit laws 8.3.7 can be adapted to yield the following statement concerning limits of functions.

8.4.7 Theorem (Limit laws for functions). *Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then*

- (a) $\lim_{x \rightarrow a} (cf)(x) = cL$
- (b) $\lim_{x \rightarrow a} (f + g)(x) = L + M$
- (c) $\lim_{x \rightarrow a} (fg)(x) = LM$
- (d) $\lim_{x \rightarrow a} (f/g)(x) = L/M$ if $M \neq 0$ and $g(x) \neq 0$ for all $x \neq a$ in an open interval containing a
- (e) $\lim_{x \rightarrow a} (f^k)(x) = L^k$

(f) $\lim_{x \rightarrow a} (f^{1/k})(x) = L^{1/k}$ if $f(x) \geq 0$ for all $x \neq a$ in an open interval containing a

The definition of limit of a function and the proof of the limit laws for functions on $\mathbb{R}$ are first steps in a rigorous study of calculus, which is called real analysis. Limits of functions are used for key definitions in real analysis: continuity (see Problem 21), the derivative, and the Riemann integral.

8.5 Problems

Problem 18 (Construction of $\mathbb{R}$). The rational numbers, $\mathbb{Q}$, can be constructed from $\mathbb{Z}$ (Problem 15). In this problem, we will use $\mathbb{Q}$ to construct the set of real numbers, $\mathbb{R}$.

A sequence (a_n) of rational numbers is **Cauchy** if for all $\epsilon > 0$, there is an integer N such that $|a_n - a_m| < \epsilon$ for all $n, m > N$. In words, this means that a_n and a_m can be made arbitrarily close together by taking n, m sufficiently large.

For instance, given an infinite list $b_1, b_2, b_3, \dots$ of digits (integers between 0 and 9), consider the sequence (a_n) defined by using the b_i to construct decimals of increasing length:

$$(0.b_1, 0.b_1b_2, 0.b_1b_2b_3, \dots).$$

Namely $a_n = 0.b_1b_2 \dots b_n$ is a decimal with n decimal digits. Since finite decimals are rational, (a_n) is a sequence of rational numbers.

(a) Prove that the sequence (a_n) defined by $a_n = 0.b_1b_2 \dots b_n$ is Cauchy.

Let R denote the set of all Cauchy sequences of rational numbers. Consider the relation $\sim$ on R given by $(a_n) \sim (b_n)$ if for all rational $\epsilon > 0$, there is an integer N such that $|a_n - b_n| < \epsilon$ for all $n > N$. (In words, $(a_n) \sim (b_n)$ means the difference between a_n and b_n can be made arbitrarily close to 0 by taking n sufficiently large.)

(b) Prove that $\sim$ is an equivalence relation.

(Hint: for transitivity, use the triangle inequality 8.1.3.)

The set of real numbers is defined as the quotient set $\mathbb{R} = R/\sim$. Thus, a real number is an equivalence class $[(a_n)]$, where (a_n) is a Cauchy sequence of rational numbers. The idea is that $\mathbb{R}$ should be viewed as the set of all possible limits of convergent sequences of rational numbers. One can prove that every Cauchy sequence converges to a real number, and $(a_n) \sim (b_n)$ if and only if $\lim a_n = \lim b_n$. Thus the quotient map $\pi: R \rightarrow \mathbb{R}$ given by $(a_n) \mapsto [(a_n)]$ can be interpreted as the function that maps (a_n) to the real number $\lim a_n$.

With this definition of $\mathbb{R}$ as equivalence classes of Cauchy sequences, one can define operations of addition and multiplication of real numbers using addition and multiplication of Cauchy sequences. Then one can prove that these operations satisfy all of the properties mentioned in 1.1.11.

Problem 19. Prove that if the sequence (a_n) converges, then it has a unique limit.

Problem 20.

- (a) Write a precise statement involving L , ϵ , and δ that articulates the meaning of “ $\lim_{x \rightarrow a} f(x)$ does not exist”.

(Hint: this is somewhat similar to the discussion in 8.2.7. The negation of a “for all” statement is a “there exists” statement, and vice versa.)

- (b) Prove that $\lim_{x \rightarrow 0} 1/x$ does not exist.

Problem 21. Let $f: D \rightarrow \mathbb{R}$ be a function, where the domain D is a subset of $\mathbb{R}$. We say that f is **continuous** at $a \in \mathbb{R}$ if for every $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(x) - f(a)| < \epsilon \text{ for all } x \text{ such that } |x - a| < \delta.$$

- (a) Prove that the function $h: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$h(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous at 0.

(Hint: use an argument similar to 8.4.5.)

- (b) Let $g: D \rightarrow \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be functions. Prove that if g is continuous at $a \in D$ and f is continuous at $g(a)$, then $f \circ g$ is continuous at a .

Note that f is continuous at a if and only if $\lim_{x \rightarrow a} f(x) = f(a)$, so the latter could also be used as a definition of “continuous at a ”. The original definition is slightly more convenient, as we can drop the condition that $x \neq a$ in the limit (do you see why?).